

2501.06662: formula evidence

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Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
2501.06662_F00001	1	■ 1.000	t	t	the t -logarithmic (Tsallis) entropies of the next-token probability distributions associ-
2501.06662_F00002	1	■ 1.000	$\pi(y \mid x) \in [0, 1]$	$\pi(y \mid x) \in [0, 1]$	enrichment of this category over the unit interval by assigning a value $\pi(y x) \in [0, 1]$ to
2501.06662_F00003	1	■ 1.000	x	x	each pair of strings x and y , which can intuitively be thought of as the probability that
2501.06662_F00004	1	■ 1.000	y	y	each pair of strings x and y , which can intuitively be thought of as the probability that
2501.06662_F00005	2	■ 1.000	$\pi(y \mid x)$	$\pi(y \mid x)$	While an explicit construction of $\pi(y x)$ was not given in [Bradley et al., 2022], we
2501.06662_F00006	2	■ 1.000	n	n	Martin, 2025, Ch. 11]. The simplest causal LMs are n -gram models, which make their
2501.06662_F00007	2	■ 1.000	(n-1)	$(n - 1)$	prediction based on the last $\textcolor{red}{(n - 1)}$ -tokens of the prompt. The relatively recent large
2501.06662_F00008	2	■ 0.528	$\dagger$	$\dagger$	<i>beginning</i> - and <i>end-of-sentence</i> tokens, which we will denote by $\perp$ and $\dagger$, respectively.
2501.06662_F00009	2	■ 1.000	a	a	So in practice, for a tokenized user input a , the model is fed the prompt $x = \perp a$ and
2501.06662_F00010	2	■ 1.000	$x = \perp a$	$x = \perp a$	So in practice, for a tokenized user input a , the model is fed the prompt $x = \perp a$ and
2501.06662_F00011	2	■ 1.000	p_x	p_x	generates a probability distribution p_x on its set of tokens. It then samples from that
2501.06662_F00012	2	■ 0.844	a_1	a_1	distribution to select a token a_1 . If a_1 is the end-of-sentence token $\dagger$, the model outputs a
2501.06662_F00013	2	■ 1.000	$x a_1$	$x a_1$	to the user. Otherwise, the string $x a_1$ is fed back into the model, which again generates
2501.06662_F00014	2	■ 1.000	p_{xa}	p_{xa}	a probability distribution p_{xa} on its set of tokens. It samples from that distribution to
2501.06662_F00015	2	■ 1.000	a_2	a_2	select a token a_2 , and the process repeats. If $a_2 = \dagger$, the model outputs aa_1 to the user.
2501.06662_F00016	2	■ 1.000	$a_2 = \dagger$	$a_2 = \dagger$	select a token a_2 , and the process repeats. If $a_2 = \dagger$, the model outputs aa_1 to the user.
2501.06662_F00017	2	■ 1.000	aa_1	aa_1	select a token a_2 , and the process repeats. If $a_2 = \dagger$, the model outputs aa_1 to the user.
2501.06662_F00018	2	■ 1.000	π	π	values from π indeed form the hom-objects of a category enriched over the unit interval.
2501.06662_F00019	3	■ 1.000	$\pi(x \mid x) = 1$	$\pi(x \mid x) = 1$	although the $\pi(y x)$ are not <i>literally</i> probabilities (for instance, we will see that $\pi(x x) = 1$
2501.06662_F00020	3	■ 1.000	$\pi(- \mid x)$	$\pi(- \mid x)$	for each object x), for each unfinished text x the function $\pi(- x)$ can be regarded as a
2501.06662_F00021	3	■ 1.000	$T(x)$	$T(x)$	probability mass function on the set $T(x)$ of <i>terminating states</i> of a prompt x , which is the
2501.06662_F00022	3	■ 0.999	$[0, \infty]$	$[0, \infty]$	$[0, \infty]$ by defining the hom-object between strings x and y to be $-\ln \pi(y x)$. Lawvere
2501.06662_F00023	3	■ 0.999	$-\ln \pi(y \mid x)$	$-\ln \pi(y \mid x)$	$[0, \infty]$ by defining the hom-object between strings x and y to be $-\ln \pi(y x)$. Lawvere
2501.06662_F00024	3	■ 0.999	$([0, \infty], \geq, +, 0)$	$([0, \infty], \geq, +, 0)$	[1973] observed that one can see a $\textcolor{red}{([0, \infty], \geq, +, 0)}$ -enriched category as a <i>generalized</i>
2501.06662_F00025	3	■ 1.000	$\mathcal{M}$	$\mathcal{M}$	In more detail, we study the generalized metric space $\mathcal{M}$ whose elements are strings
2501.06662_F00026	3	■ 1.000	$x, y, \dots$	$x, y, \dots$	$x, y, \dots$ and whose distances are given by $d(x, y) = -\ln \pi(y x)$. We use the approach
2501.06662_F00027	3	■ 1.000	$d(x, y) = -\ln \pi(y \mid x)$	$d(x, y) = -\ln \pi(y \mid x)$	$x, y, \dots$ and whose distances are given by $d(x, y) = -\ln \pi(y x)$. We use the approach
2501.06662_F00028	3	■ 1.000	$\operatorname{Mag}(t \mathcal{M})$	$\operatorname{Mag}(t \mathcal{M})$	to calculate the <i>magnitude function</i> $\operatorname{Mag}(t \mathcal{M})$, for $t > 0$.
2501.06662_F00029	3	■ 1.000	$t > 0$	$t > 0$	to calculate the <i>magnitude function</i> $\operatorname{Mag}(t \mathcal{M})$, for $\textcolor{red}{t > 0}$.
2501.06662_F00030	3	■ 0.984	$\#(T(\perp))$	$\#(T(\perp))$	1. $\textcolor{red}{\#(T(\perp))}$ is the number of possible outputs of the model,
2501.06662_F00031	4	■ 1.000	H_t	H_t	3. H_t is the t -logarithmic entropy: $H_t(p_1, \dots, p_m) = (1 - \sum_{i=1}^m p_i^t)/(t - 1)$.
2501.06662_F00032	4	■ 1.000	$H_t(p_1, \dots, p_m) = \left(1 - \sum_{i=1}^m p_i^t\right)/(t - 1)$	$H_t(p_1, \dots, p_m) = \left(1 - \sum_{i=1}^m p_i^t\right)/(t - 1)$	3. H_t is the t -logarithmic entropy: $\textcolor{red}{H_t(p_1, \dots, p_m) = (1 - \sum_{i=1}^m p_i^t)/(t - 1)}$.
2501.06662_F00033	4	■ 1.000	$(H_t)_{t \in (0, \infty) \setminus \{1\}}$	$(H_t)_{t \in (0, \infty) \setminus \{1\}}$	The entropies $\textcolor{red}{(H_t)_{t \in (0, \infty) \setminus \{1\}}}$ were first introduced by Havrda and Charvát [1967] and
2501.06662_F00034	4	■ 1.000	$t > 1$	$t > 1$	Charvát, 1967]. It follows that, when $t > 1$, the function $\operatorname{Mag}(t \mathcal{M})$ is lower bounded
2501.06662_F00035	4	■ 1.000	$H_t(p_x)$	$H_t(p_x)$	set; in this case the t -logarithmic entropy $H_t(p_x)$ is equal to $(n^{1-t} - 1)/(1 - t)$, where n
2501.06662_F00036	4	■ 1.000	$(n^{1-t} - 1)/(1 - t)$	$(n^{1-t} - 1)/(1 - t)$	set; in this case the t -logarithmic entropy $H_t(p_x)$ is equal to $(n^{1-t} - 1)/(1 - t)$, where n
2501.06662_F00037	4	■ 1.000	$t \rightarrow \infty$	$t \rightarrow \infty$	is the size of the token set. In the limit $t \rightarrow \infty$, these bounds take the simple form
2501.06662_F00038	4	■ 1.000	$\mathcal{M}_1$	$\mathcal{M}_1$	language models with associated spaces $\mathcal{M}_1$ and $\mathcal{M}_2$ assign the same next-token proba-
2501.06662_F00039	4	■ 1.000	$\mathcal{M}_2$	$\mathcal{M}_2$	language models with associated spaces $\mathcal{M}_1$ and $\mathcal{M}_2$ assign the same next-token proba-
2501.06662_F00040	4	■ 0.405	$p_{\bullet}^{(1)}$	$p_{\bullet}^{(1)}$	bilities (respectively, $p_{\bullet}^{(1)}$ and $p_{\bullet}^{(2)}$) to all prompts but one, called x' , in such a way that
2501.06662_F00041	4	■ 0.405	$p_{\bullet}^{(2)}$	$p_{\bullet}^{(2)}$	bilities (respectively, $p_{\bullet}^{(1)}$ and $p_{\bullet}^{(2)}$) to all prompts but one, called x' , in such a way that
2501.06662_F00042	4	■ 0.405	x'	x'	bilities (respectively, $p_{\bullet}^{(1)}$ and $p_{\bullet}^{(2)}$) to all prompts but one, called x' , in such a way that

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2501.06662_F00043	4	■ 1.000	$H_{\{t\}\left(p_{x^{\prime}}^{\{1\}}\right)}>H_{\{t\}\left(p_x^{\{2\}}\right)}$	$H_t\left(p_{x^{\prime}}^{\{1\}}\right)>H_t\left(p_x^{\{2\}}\right)$	$H_t(p_{x^{\prime}}^{\{1\}})>H_t(p_x^{\{2\}})$, then $\text{Mag}(t\mathcal{M}_1)>\text{Mag}(t\mathcal{M}_2)$.
2501.06662_F00044	4	■ 1.000	$\operatorname{Mag}\left(t\ \mathcal{M}_{\{1\}}\right)>\operatorname{Mag}\left(t\ \mathcal{M}_{\{2\}}\right)$	$\text{Mag}(t\mathcal{M}_1)>\text{Mag}(t\mathcal{M}_2)$	$H_t(p_{x^{\prime}}^{\{1\}})>H_t(p_x^{\{2\}})$, then $\boxed{\text{Mag}(t\mathcal{M}_1)>\text{Mag}(t\mathcal{M}_2)}$.
2501.06662_F00045	4	■ 1.000	$t<1$	$t<1$	The situation is less clear when $t<1$. For instance, for a maximally random LM,
2501.06662_F00046	4	■ 0.999	$\lim_{t\rightarrow 0}\operatorname{Mag}(t\ \mathcal{M})=(1-n)\cdot\#\left(\operatorname{ob}(\mathcal{M})\backslash T(\perp)\right)\backslash T(\perp)$	$\lim_{t\rightarrow 0}\text{Mag}(t\mathcal{M})=(1-n)\cdot\#(\text{ob}(\mathcal{M})\backslash T(\perp))+\#(T(\perp))$	we have $\lim_{t\rightarrow 0}\boxed{\text{Mag}(t\mathcal{M})}=(1-n)\cdot\#(\text{ob}(\mathcal{M})\backslash T(\perp))+\#(T(\perp))$, which is negative
2501.06662_F00047	4	■ 0.696	$n>1$	$n>1$	(since $\boxed{n>1}$, unless $\dagger$ is the only token besides $\perp$). However, even for metric spaces, this
2501.06662_F00048	4	■ 0.999	$t=1$	$t=1$	We shall see that the derivative of $\text{Mag}(t\mathcal{M})$ at $\boxed{t=1}$ is a sum of Shannon entropies:
2501.06662_F00049	4	■ 0.896	$\sum_{x\in\operatorname{ob}(\mathcal{M})\backslash T(\perp)}H\left(p_x\right)$	$\sum_{x\in\text{ob}(\mathcal{M})\backslash T(\perp)}H(p_x)$	$\sum_{x\in\boxed{\text{ob}(\mathcal{M})\backslash T(\perp)}}\boxed{H(p_x)}$. Analogous results hold for the subspace $\mathcal{M}_x$ of texts that extend a
2501.06662_F00050	4	■ 0.896	$\mathcal{M}_{\{x\}}$	$\mathcal{M}_x$	$\sum_{x\in\text{ob}(\mathcal{M})\backslash T(\perp)}H(p_x)$. Analogous results hold for the subspace $\mathcal{M}_x$ of texts that extend a
2501.06662_F00051	4	■ 0.985	$\left(\sum_y\mu(x,y)\right)_x$	$\left(\sum_y\mu(x,y)\right)_x$	context of metric spaces, <i>weighting vectors</i> [Leinster, 2013]—such as $(\sum_y\mu(x,y))_x$ when
2501.06662_F00052	5	■ 1.000	μ	μ	the Möbius coefficients μ exist—effectively detect the boundary of certain subsets of Eu-
2501.06662_F00053	5	■ 1.000	ℓ	ℓ	where the first sum ranges over those ℓ that may appear as finite lengths of paths in $\mathcal{M}$.
2501.06662_F00054	5	■ 0.985	$\mathcal{L}$	$\mathcal{L}$	simple case. Subsection 2.15 shows that π defines a $[0,1]$ -category $\mathcal{L}$ of texts in the sense
2501.06662_F00055	5	■ 0.977	$-\ln:[0,1]\rightarrow[0,\infty]$	$-\ln:[0,1]\rightarrow[0,\infty]$	space $\mathcal{M}$ using the isomorphism of categories $\boxed{-\ln:[0,1]\rightarrow[0,\infty]}$.
2501.06662_F00056	6	■ 1.000	A	A	2.1. TOKENS AND TEXTS. Let A be a finite alphabet (a set of tokens) and consider all
2501.06662_F00057	6	■ 1.000	$a,b,\ldots$	$a,b,\ldots$	finite strings $\boxed{a,b,\ldots}$ from the free monoid A^* . In LMs, tokens can correspond to words,
2501.06662_F00058	6	■ 1.000	$A^{\ast}$	A^*	finite strings $a,b,\ldots$ from the free monoid A^* . In LMs, tokens can correspond to words,
2501.06662_F00059	6	■ 0.999	$a\leq b$	$a\leq b$	elements of A^* <i>strings</i> or <i>texts</i> . Write $a\leq b$ whenever a is a <i>prefix</i> of b , that is, whenever
2501.06662_F00060	6	■ 0.999	b	b	elements of A^* <i>strings</i> or <i>texts</i> . Write $a\leq b$ whenever a is a <i>prefix</i> of b , that is, whenever
2501.06662_F00061	6	■ 1.000	$b=a\ a^{\prime}$	$b=aa'$	$\boxed{b=aa'}$ for some $a'\in A^*$. This defines a partial order since $\leq$ is reflexive (as A^* contains
2501.06662_F00062	6	■ 1.000	$a^{\prime}\in A^{\ast}$	$a'\in A^*$	$b=aa'$ for some $\boxed{a'\in A^*}$. This defines a partial order since $\leq$ is reflexive (as A^* contains
2501.06662_F00063	6	■ 1.000	$\leq$	$\leq$	$b=aa'$ for some $a'\in A^*$. This defines a partial order since $\leq$ is reflexive (as A^* contains
2501.06662_F00064	6	■ 1.000	ϵ	ϵ	the empty string ϵ), transitive, and antisymmetric. We denote by $ a $ the length of a string
2501.06662_F00065	6	■ 1.000	$ a $	$ a $	the empty string ϵ), transitive, and antisymmetric. We denote by $ a $ the length of a string
2501.06662_F00066	6	■ 1.000	$a\in A^{\ast}$	$a\in A^*$	$\boxed{a\in A^*}$.
2501.06662_F00067	6	■ 1.000	$(P,\leq)$	$(P,\leq)$	We regard a partially ordered set $(P,\leq)$ as a thin category with object set P and such
2501.06662_F00068	6	■ 1.000	P	P	We regard a partially ordered set $(P,\leq)$ as a thin category with object set P and such
2501.06662_F00069	6	■ 1.000	$x\rightarrow y$	$x\rightarrow y$	that $x\rightarrow y$ if and only if $x\leq y$. Given $N\in\mathbb{N}$, we define $\mathbf{L}:=\mathbf{L}^{\leq N}$ as the full subcategory
2501.06662_F00070	6	■ 1.000	$x\leq y$	$x\leq y$	that $x\rightarrow y$ if and only if $x\leq y$. Given $N\in\mathbb{N}$, we define $\mathbf{L}:=\mathbf{L}^{\leq N}$ as the full subcategory
2501.06662_F00071	6	■ 1.000	$N\in\mathbb{N}$	$N\in\mathbb{N}$	that $x\rightarrow y$ if and only if $x\leq y$. Given $N\in\mathbb{N}$, we define $\mathbf{L}:=\mathbf{L}^{\leq N}$ as the full subcategory
2501.06662_F00072	6	■ 1.000	$\mathrm{L}:=\mathrm{L}^{\leq N}$	$\mathbf{L}:=\mathbf{L}^{\leq N}$	that $x\rightarrow y$ if and only if $x\leq y$. Given $N\in\mathbb{N}$, we define $\boxed{\mathbf{L}:=\mathbf{L}^{\leq N}}$ as the full subcategory
2501.06662_F00073	6	■ 1.000	$\left((A\cup\{\perp,\dagger\})^{\ast},\leq\right)$	$((A\cup\{\perp,\dagger\})^*,\leq)$	of $\boxed{(A\cup\{\perp,\dagger\})^*,\leq}$ made of strings that start with $\perp$ and are followed by at most $N-1$
2501.06662_F00074	6	■ 1.000	$N-1$	$N-1$	of $((A\cup\{\perp,\dagger\})^*,\leq)$ made of strings that start with $\perp$ and are followed by at most $N-1$
2501.06662_F00075	6	■ 0.803	x,y	x,y	More explicitly, given objects x,y of $\mathbf{L}$, there is a morphism $x\rightarrow y$ whenever $x\leq y$ as
2501.06662_F00076	6	■ 1.000	$(A\cup\{\perp,\dagger\})^{\ast}$	$(A\cup\{\perp,\dagger\})^*$	elements of $\boxed{(A\cup\{\perp,\dagger\})^*}$, that is, if x is a prefix of y . We refer to an object of the form
2501.06662_F00077	6	■ 0.918	$\perp a$	$\perp a$	$\perp a$ as an <i>unfinished text</i> and to an object of the form $\perp a\dagger$ as a <i>finished text</i> . For any
2501.06662_F00078	6	■ 0.918	$\perp a\dagger$	$\perp a\dagger$	$\perp a$ as an <i>unfinished text</i> and to an object of the form $\perp a\dagger$ as a <i>finished text</i> . For any
2501.06662_F00079	6	■ 0.999	$ x $	$ x $	that it has a natural “grading” given by the length $ x $ of strings x . We remark that
2501.06662_F00080	6	■ 1.000	$ \perp =1$	$ \perp =1$	$ \perp =1$.
2501.06662_F00081	6	■ 0.981	$\operatorname{ob}(\mathrm{L})$	$\text{ob}(\mathbf{L})$	It will be convenient to write the set $\text{ob}(\mathbf{L})$ as a disjoint union $\bigsqcup_{j=0}^{N-1}\mathbf{L}^{(j)}$, where $\mathbf{L}^{(j)}$
2501.06662_F00082	6	■ 0.981	$\bigsqcup_{j=0}^{N-1}\mathrm{L}^{\sim\{j\}}$	$\bigsqcup_{j=0}^{N-1}\mathbf{L}^{(j)}$	It will be convenient to write the set $\text{ob}(\mathbf{L})$ as a disjoint union $\bigsqcup_{j=0}^{N-1}\mathbf{L}^{(j)}$, where $\mathbf{L}^{(j)}$
2501.06662_F00083	6	■ 0.981	$\mathrm{L}^{\sim\{j\}}$	$\mathbf{L}^{(j)}$	It will be convenient to write the set $\text{ob}(\mathbf{L})$ as a disjoint union $\bigsqcup_{j=0}^{N-1}\mathbf{L}^{(j)}$, where $\mathbf{L}^{(j)}$
2501.06662_F00084	6	■ 0.998	$1+j$	$1+j$	consists of strings of length $1+j$. (Remark that $\mathbf{L}$ is a free category on a graph, so j
2501.06662_F00085	6	■ 0.998	j	j	consists of strings of length $1+j$. (Remark that $\mathbf{L}$ is a free category on a graph, so j
2501.06662_F00086	6	■ 0.990	$L^{\sim\{j\}}$	$L^{(j)}$	measures the graph-theoretic distance of the objects in $L^{(j)}$ to $\perp$.) Aside from identities,

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2501.06662_F00087	6	■ 0.996	$\mathrm{L}^{\{i\}}$	$\mathrm{L}^{(i)}$	the arrows of L only go from elements of $\mathrm{L}^{(i)}$ to elements of $\mathrm{L}^{(j)}$ for $j > i$. Similarly,
2501.06662_F00088	6	■ 0.996	$j > i$	$j > i$	the arrows of L only go from elements of $\mathrm{L}^{(i)}$ to elements of $\mathrm{L}^{(j)}$ for $j > i$. Similarly,
2501.06662_F00089	6	■ 0.881	L_x	L_x	given a string x in L , there is a full subcategory L_x of L whose objects are $y \in \mathrm{L}$ such
2501.06662_F00090	6	■ 0.881	$y \in \mathrm{L}$	$y \in \mathrm{L}$	given a string x in L , there is a full subcategory L_x of L whose objects are $y \in \mathrm{L}$ such
2501.06662_F00091	6	■ 0.993	$\operatorname{ob}(\mathrm{L}_x) = \bigsqcup_{j=0}^{N- x } \mathrm{L}_x^{\{j\}}$	$\operatorname{ob}(\mathrm{L}_x) = \bigsqcup_{j=0}^{N- x } \mathrm{L}_x^{(j)}$	that $x \rightarrow y$; in this case, we have $\operatorname{ob}(\mathrm{L}_x) = \bigsqcup_{j=0}^{N- x } \mathrm{L}_x^{(j)}$, where $\mathrm{L}_x^{(j)}$ consists only of those
2501.06662_F00092	6	■ 0.993	$\mathrm{L}_x^{\{j\}}$	$\mathrm{L}_x^{(j)}$	that $x \rightarrow y$; in this case, we have $\operatorname{ob}(\mathrm{L}_x) = \bigsqcup_{j=0}^{N- x } \mathrm{L}_x^{(j)}$, where $\mathrm{L}_x^{(j)}$ consists only of those
2501.06662_F00093	6	■ 1.000	$ x + j$	$ x + j$	strings of length $ x + j$, namely, those strings that extend x on the right by j tokens. So,
2501.06662_F00094	6	■ 1.000	$\mathrm{L}_x^{\{0\}} = \{x\}$	$\mathrm{L}_x^{(0)} = \{x\}$	$\mathrm{L}_x^{(0)} = \{x\}$ and $\mathrm{L}_x^{(1)} = \{xa_1 : a_1 \in A \cup \{\dagger\}\}$ and $\mathrm{L}_x^{(2)} = \{xa_1a_2 : a_1 \in A, a_2 \in A \cup \{\dagger\}\}$ and
2501.06662_F00095	6	■ 1.000	$\mathrm{L}_x^{\{1\}} = \{xa_1 : a_1 \in A \cup \{\dagger\}\}$	$\mathrm{L}_x^{(1)} = \{xa_1 : a_1 \in A \cup \{\dagger\}\}$	$\mathrm{L}_x^{(0)} = \{x\}$ and $\mathrm{L}_x^{(1)} = \{xa_1 : a_1 \in A \cup \{\dagger\}\}$ and $\mathrm{L}_x^{(2)} = \{xa_1a_2 : a_1 \in A, a_2 \in A \cup \{\dagger\}\}$ and
2501.06662_F00096	6	■ 1.000	$\mathrm{L}_x^{\{2\}} = \{xa_1a_2 : a_1 \in A, a_2 \in A \cup \{\dagger\}\}$	$\mathrm{L}_x^{(2)} = \{xa_1a_2 : a_1 \in A, a_2 \in A \cup \{\dagger\}\}$	$\mathrm{L}_x^{(0)} = \{x\}$ and $\mathrm{L}_x^{(1)} = \{xa_1 : a_1 \in A \cup \{\dagger\}\}$ and $\mathrm{L}_x^{(2)} = \{xa_1a_2 : a_1 \in A, a_2 \in A \cup \{\dagger\}\}$ and
2501.06662_F00097	6	■ 1.000	$\mathrm{L}_{\perp}^{\{j\}} = \mathrm{L}^{\{j\}}$	$\mathrm{L}_{\perp}^{(j)} = \mathrm{L}^{(j)}$	so on. Remark that $\mathrm{L}_{\perp}^{(j)} = \mathrm{L}^{(j)}$ for any $j \geq 0$, and if x is a finished text, then L_x has a
2501.06662_F00098	6	■ 1.000	$j \geq 0$	$j \geq 0$	so on. Remark that $\mathrm{L}_{\perp}^{(j)} = \mathrm{L}^{(j)}$ for any $j \geq 0$, and if x is a finished text, then L_x has a
2501.06662_F00099	6	■ 0.843	$\top$	$\top$	of $\top$, given our use of $\perp$ for the beginning-of-sentence symbol. To start, the category L does not have a
2501.06662_F00100	6	■ 0.843	$\mathbf{L}$	$\mathbf{L}$	of $\top$, given our use of $\perp$ for the beginning-of-sentence symbol. To start, the category L does not have a
2501.06662_F00101	7	■ 1.000	$p_x := p(- \mid x) : A \cup \{\dagger\} \rightarrow [0, 1]$	$p_x := p(- \mid x) : A \cup \{\dagger\} \rightarrow [0, 1]$	$p_x := p(- \mid x) : A \cup \{\dagger\} \rightarrow [0, 1]$. After generating p_x , the model then samples a token
2501.06662_F00102	7	■ 1.000	$a_1 = \dagger$	$a_1 = \dagger$	a_1 according to p_x . If $a_1 = \dagger$, then the process terminates and the model outputs a .
2501.06662_F00103	7	■ 1.000	$p_{\perp aa_1}$	$p_{\perp aa_1}$	generates a probability distribution $p_{\perp aa_1}$ on $A \cup \{\dagger\}$, samples a token a_2 according to
2501.06662_F00104	7	■ 1.000	$A \cup \{\dagger\}$	$A \cup \{\dagger\}$	generates a probability distribution $p_{\perp aa_1}$ on $A \cup \{\dagger\}$, samples a token a_2 according to
2501.06662_F00105	7	■ 0.827	$p_{\perp aa_1 a_2}$	$p_{\perp aa_1 a_2}$	generate $p_{\perp aa_1 a_2}$, and so on.
2501.06662_F00106	7	■ 1.000	$y = xa_1 \cdots a_{N- x }$	$y = xa_1 \cdots a_{N- x }$	$y = xa_1 \cdots a_{N- x }$ reaches a maximum length N ; we call N the <i>cutoff</i> . We can think of
2501.06662_F00107	7	■ 1.000	N	N	$y = xa_1 \cdots a_{N- x }$ reaches a maximum length N ; we call N the <i>cutoff</i> . We can think of
2501.06662_F00108	7	■ 1.000	$a = a(x) \in A \cup \{\dagger\}$	$a = a(x) \in A \cup \{\dagger\}$	x , there is $a = a(x) \in A \cup \{\dagger\}$ such that $p_x(a) = 1$ (and therefore $p_x(a') = 0$ for any
2501.06662_F00109	7	■ 1.000	$p_x(a) = 1$	$p_x(a) = 1$	x , there is $a = a(x) \in A \cup \{\dagger\}$ such that $p_x(a) = 1$ (and therefore $p_x(a') = 0$ for any
2501.06662_F00110	7	■ 1.000	$p_x(a') = 0$	$p_x(a') = 0$	x , there is $a = a(x) \in A \cup \{\dagger\}$ such that $p_x(a) = 1$ (and therefore $p_x(a') = 0$ for any
2501.06662_F00111	7	■ 1.000	$a' \in (A \cup \{\dagger\}) \setminus \{a\}$	$a' \in (A \cup \{\dagger\}) \setminus \{a\}$	$a' \in (A \cup \{\dagger\}) \setminus \{a\}$. Although this example is quite artificial, it will appear as an extreme
2501.06662_F00112	7	■ 0.720	$p_{\perp a_1 \cdots a_n}(-)$	$p_{\perp a_1 \cdots a_n}(-)$	2.4. EXAMPLE. If $p_{\perp a_1 \cdots a_n}(-)$ only depends on a_n for any $n \geq 1$, then the LM corresponds
2501.06662_F00113	7	■ 0.720	a_n	a_n	2.4. EXAMPLE. If $p_{\perp a_1 \cdots a_n}(-)$ only depends on a_n for any $n \geq 1$, then the LM corresponds
2501.06662_F00114	7	■ 0.720	$n \geq 1$	$n \geq 1$	2.4. EXAMPLE. If $p_{\perp a_1 \cdots a_n}(-)$ only depends on a_n for any $n \geq 1$, then the LM corresponds
2501.06662_F00115	7	■ 1.000	$\tilde{A} = A \cup \{\dagger\}$	$\tilde{A} = A \cup \{\dagger\}$	2020, Ch. 6]), $\tilde{A} = A \cup \{\dagger\}$ is the set of states and $\dagger$ is an absorbing state. The numbers
2501.06662_F00116	7	■ 1.000	$P_n(a_n, a_{n+1}) := p_{\perp a_1 \cdots a_n}(a_{n+1})$	$P_n(a_n, a_{n+1}) := p_{\perp a_1 \cdots a_n}(a_{n+1})$	$P_n(a_n, a_{n+1}) := p_{\perp a_1 \cdots a_n}(a_{n+1})$ form a matrix $P_n : \tilde{A} \times \tilde{A} \rightarrow [0, 1]$, which has positive
2501.06662_F00117	7	■ 1.000	$P_n : \tilde{A} \times \tilde{A} \rightarrow [0, 1]$	$P_n : \tilde{A} \times \tilde{A} \rightarrow [0, 1]$	$P_n(a_n, a_{n+1}) := p_{\perp a_1 \cdots a_n}(a_{n+1})$ form a matrix $P_n : \tilde{A} \times \tilde{A} \rightarrow [0, 1]$, which has positive
2501.06662_F00118	7	■ 0.998	$\sum_{a' \in \tilde{A}} P_n(a, a') = 1$	$\sum_{a' \in \tilde{A}} P_n(a, a') = 1$	entries and is <i>stochastic</i> , meaning that each row sums to 1, i.e. $\sum_{a' \in \tilde{A}} P_n(a, a') = 1$. By
2501.06662_F00119	7	■ 0.971	$P_n(\dagger, \dagger) = 1$	$P_n(\dagger, \dagger) = 1$	convention we are assuming here that $P_n(\dagger, \dagger) = 1$ and $P_n(\dagger, a) = 0$ for any $a \in A$, which
2501.06662_F00120	7	■ 0.971	$P_n(\dagger, a) = 0$	$P_n(\dagger, a) = 0$	convention we are assuming here that $P_n(\dagger, \dagger) = 1$ and $P_n(\dagger, a) = 0$ for any $a \in A$, which
2501.06662_F00121	7	■ 0.971	$a \in A$	$a \in A$	convention we are assuming here that $P_n(\dagger, \dagger) = 1$ and $P_n(\dagger, a) = 0$ for any $a \in A$, which
2501.06662_F00122	7	■ 1.000	$n > 2$	$n > 2$	connection with homogeneous and irreducible Markov chains. If for all $n > 2$ it holds
2501.06662_F00123	7	■ 1.000	$P_n = P_1$	$P_n = P_1$	that $P_n = P_1$, then the chain is said to be <i>homogeneous</i> . A Markov chain is <i>irreducible</i>
2501.06662_F00124	7	■ 1.000	$a' \in A$	$a' \in A$	if there is a nontrivial probability of going from any state $a \in A$ to any state $a' \in A$ in
2501.06662_F00125	7	■ 1.000	$P_n(a, \dagger) = 0$	$P_n(a, \dagger) = 0$	if $P_n(a, \dagger) = 0$ for every $n \geq 1$ and $a \in A$, then $P_n _{A \times A}$ is also stochastic and $(P_n)_{n \geq 1}$
2501.06662_F00126	7	■ 1.000	$P_n _{A \times A}$	$P_n _{A \times A}$	if $P_n(a, \dagger) = 0$ for every $n \geq 1$ and $a \in A$, then $P_n _{A \times A}$ is also stochastic and $(P_n)_{n \geq 1}$
2501.06662_F00127	7	■ 1.000	$(P_n)_{n \geq 1}$	$(P_n)_{n \geq 1}$	if $P_n(a, \dagger) = 0$ for every $n \geq 1$ and $a \in A$, then $P_n _{A \times A}$ is also stochastic and $(P_n)_{n \geq 1}$
2501.06662_F00128	7	■ 1.000	$q : A \rightarrow [0, 1]$	$q : A \rightarrow [0, 1]$	and Stirzaker, 2020, Sec. 6.6]), which is a probability vector $q : A \rightarrow [0, 1]$ such that
2501.06662_F00129	7	■ 1.000	$\sum_{a' \in A} q(a)P(a, a') = q(a')$	$\sum_{a' \in A} q(a)P(a, a') = q(a')$	$\sum_{a' \in A} q(a)P(a, a') = q(a')$. In this setting, we will see in Section 4 that if additionally
2501.06662_F00130	7	■ 1.000	$p_{\perp}(a) = q(a)$	$p_{\perp}(a) = q(a)$	$p_{\perp}(a) = q(a)$, then we can recover the Kolmogorov–Sinai entropy of the Markov chain
2501.06662_F00131	8	■ 1.000	$n - 1$	$n - 1$	2.5. EXAMPLE. An <i>n-gram model</i> is a Markov chain “with memory” of order $n - 1$, in

Identifier	Page	Conf.	LaTeX source	Rendered	Image
2501.06662_F00132	8	■ 0.522	$p_{\perp a_1 \cdots a_i}$	$p_{\perp a_1 \cdots a_i}$	the sense that $p_{\perp a_1 \cdots a_i}(-)$ only depends on the last $n - 1$ tokens of $\perp a_1 \cdots a_i$ (or all of
2501.06662_F00133	8	■ 0.522	$\perp a_1 \cdots a_i$	$\perp a_1 \cdots a_i$	the sense that $p_{\perp a_1 \cdots a_i}(-)$ only depends on the last $n - 1$ tokens of $\perp a_1 \cdots a_i$ (or all of
2501.06662_F00134	8	■ 1.000	$i < n - 2$	$i < n - 2$	them if $i < n - 2$). For details see [Jurafsky and Martin, 2025, Ch. 3].
2501.06662_F00135	8	■ 1.000	$p^{(\theta)} : A^* \rightarrow \Delta(A), \quad \text{\texttt{\textbackslash mapsto}} p_{\perp a}^{(\theta)}(-)$	$p^{(\theta)} : A^* \rightarrow \Delta(A), \quad a \mapsto p_{\perp a}^{(\theta)}(-)$	function $p^{(\theta)} : A^* \rightarrow \Delta(A), \quad a \mapsto p_{\perp a}^{(\theta)}(-)$. Here $\Delta(A)$ denotes the set of probability mass
2501.06662_F00136	8	■ 1.000	$\Delta(A)$	$\Delta(A)$	function $p^{(\theta)} : A^* \rightarrow \Delta(A), \quad a \mapsto p_{\perp a}^{(\theta)}(-)$. Here $\Delta(A)$ denotes the set of probability mass
2501.06662_F00137	8	■ 1.000	θ	θ	functions on A , and θ the network’s parameters (connection weights between neurons) and
2501.06662_F00138	8	■ 0.998	$\left(a_1, \ldots, a_{n-1}\right)$	$(a_1, \ldots, a_{n-1})$	to predict as well as possible the last token a_n from $(a_1, \ldots, a_{n-1})$, given any tokenized
2501.06662_F00139	8	■ 0.998	$\left(a_1, \ldots, a_n\right)$	$(a_1, \ldots, a_n)$	fragment of text $(a_1, \ldots, a_n)$ in a corpus $C \subset A^*$. For details see [Jurafsky and Martin,
2501.06662_F00140	8	■ 0.998	$C \subset A^*$	$C \subset A^*$	fragment of text $(a_1, \ldots, a_n)$ in a corpus $C \subset A^*$. For details see [Jurafsky and Martin,
2501.06662_F00141	8	■ 1.000	$y = x a^{\prime}$	$y = x a'$	In the first case, $y = x a'$ for some $a' \in A^*$ such that $ a' = N - x $; for such a
2501.06662_F00142	8	■ 1.000	$ a^{\prime} = N - x $	$ a' = N - x $	In the first case, $y = x a'$ for some $a' \in A^*$ such that $ a' = N - x $; for such a
2501.06662_F00143	8	■ 1.000	$a a^{\prime}$	aa'	terminating state the model outputs aa' to the user according to some probability dis-
2501.06662_F00144	8	■ 1.000	$y = x a^{\prime \prime} \dagger$	$y = x a'' \dagger$	tribution. In the second case, $y = x a'' \dagger$ for some $a'' \in A^*$ such that $ a'' \leq N - x - 1$;
2501.06662_F00145	8	■ 1.000	$a^{\prime \prime} \in A^*$	$a'' \in A^*$	tribution. In the second case, $y = x a'' \dagger$ for some $a'' \in A^*$ such that $ a'' \leq N - x - 1$;
2501.06662_F00146	8	■ 1.000	$ a^{\prime \prime} \leq N - x - 1$	$ a'' \leq N - x - 1$	tribution. In the second case, $y = x a'' \dagger$ for some $a'' \in A^*$ such that $ a'' \leq N - x - 1$;
2501.06662_F00147	8	■ 1.000	$a a^{\prime \prime}$	aa''	for such a terminating state the model outputs aa'' according to some probability distri-
2501.06662_F00148	8	■ 1.000	$N > 5$	$N > 5$	production. For example, when $N > 5$, the probability of generating the output $b =$
2501.06662_F00149	8	■ 1.000	$b =$	$b =$	production. For example, when $N > 5$, the probability of generating the output $b =$
2501.06662_F00150	8	■ 1.000	$a_1 a_2 a_3 a_4 a_5 \in A^*$	$a_1 a_2 a_3 a_4 a_5 \in A^*$	$a_1 a_2 a_3 a_4 a_5 \in A^*$ given the user’s input $a = a_1 a_2$ would be
2501.06662_F00151	8	■ 1.000	$a = a_1 a_2$	$a = a_1 a_2$	$a_1 a_2 a_3 a_4 a_5 \in A^*$ given the user’s input $a = a_1 a_2$ would be
2501.06662_F00152	8	■ 1.000	$f(x)$	$f(x)$	a prompt x of length greater than $N - 1$, one needs to produce first a subrogate prompt $f(x)$ such that
2501.06662_F00153	8	■ 1.000	$ f(x) \leq N - 1$	$ f(x) \leq N - 1$	$ f(x) \leq N - 1$ and feed it to the language model to generate $p_{f(x)}$ and sample a token.
2501.06662_F00154	8	■ 1.000	$p_{f(x)}$	$p_{f(x)}$	$ f(x) \leq N - 1$ and feed it to the language model to generate $p_{f(x)}$ and sample a token.
2501.06662_F00155	9	■ 1.000	$x = \perp a_1 \cdots a_t$	$x = \perp a_1 \cdots a_t$	In the third case, we assume that $x = \perp a_1 \cdots a_t$ and $y = x a_{t+1} \cdots a_{t+k}$ for some $k \geq 1$,
2501.06662_F00156	9	■ 1.000	$y = x a_{t+1} \cdots a_{t+k}$	$y = x a_{t+1} \cdots a_{t+k}$	In the third case, we assume that $x = \perp a_1 \cdots a_t$ and $y = x a_{t+1} \cdots a_{t+k}$ for some $k \geq 1$,
2501.06662_F00157	9	■ 1.000	$k \geq 1$	$k \geq 1$	In the third case, we assume that $x = \perp a_1 \cdots a_t$ and $y = x a_{t+1} \cdots a_{t+k}$ for some $k \geq 1$,
2501.06662_F00158	9	■ 0.977	$\left(a_i\right)_{i=1}^{t+k-1} \subset A$	$(a_i)_{i=1}^{t+k-1} \subset A$	$(a_i)_{i=1}^{t+k-1} \subset A$, and $a_{t+k} \in A \cup \{\dagger\}$. The symbol $y_{<t+i}$ denotes the string $\perp a_1 \cdots a_{t+i-1}$,
2501.06662_F00159	9	■ 0.977	$a_{t+k} \in A \cup \{\dagger\}$	$a_{t+k} \in A \cup \{\dagger\}$	$(a_i)_{i=1}^{t+k-1} \subset A$, and $a_{t+k} \in A \cup \{\dagger\}$. The symbol $y_{<t+i}$ denotes the string $\perp a_1 \cdots a_{t+i-1}$,
2501.06662_F00160	9	■ 0.977	$y_{<t+i}$	$y_{<t+i}$	$(a_i)_{i=1}^{t+k-1} \subset A$, and $a_{t+k} \in A \cup \{\dagger\}$. The symbol $y_{<t+i}$ denotes the string $\perp a_1 \cdots a_{t+i-1}$,
2501.06662_F00161	9	■ 0.977	$\perp a_1 \cdots a_{t+i-1}$	$\perp a_1 \cdots a_{t+i-1}$	$(a_i)_{i=1}^{t+k-1} \subset A$, and $a_{t+k} \in A \cup \{\dagger\}$. The symbol $y_{<t+i}$ denotes the string $\perp a_1 \cdots a_{t+i-1}$,
2501.06662_F00162	9	■ 1.000	$y_{<t+1} = x$	$y_{<t+1} = x$	and $y_{<t+1} = x$.
2501.06662_F00163	9	■ 1.000	$m = 0$	$m = 0$	PROOF. If $m = 0$, then $T(x) = \{x\}$ and by definition $\pi(x x) = 1$. If $m = 1$, then
2501.06662_F00164	9	■ 1.000	$T(x) = \{x\}$	$T(x) = \{x\}$	PROOF. If $m = 0$, then $T(x) = \{x\}$ and by definition $\pi(x x) = 1$. If $m = 1$, then
2501.06662_F00165	9	■ 1.000	$m = 1$	$m = 1$	PROOF. If $m = 0$, then $T(x) = \{x\}$ and by definition $\pi(x x) = 1$. If $m = 1$, then
2501.06662_F00166	9	■ 1.000	$T(x) = \{x a \mid a \in A \cup \{\dagger\}\}$	$T(x) = \{x a \mid a \in A \cup \{\dagger\}\}$	$T(x) = \{x a \mid a \in A \cup \{\dagger\}\}$ and
2501.06662_F00167	9	■ 1.000	$m \geq 1$	$m \geq 1$	since p_x is assumed to be a probability mass function on $A \cup \{\dagger\}$. For general $m \geq 1$,
2501.06662_F00168	9	■ 1.000	$a \in A^m$	$a \in A^m$	texts of length N . We obtain (3) by identifying $a \in A^m$ with $(a', a'') \in A^{m-1} \times A$.
2501.06662_F00169	9	■ 1.000	$\left(a^{\prime}, a^{\prime \prime}\right) \in A^{m-1} \times A$	$(a', a'') \in A^{m-1} \times A$	texts of length N . We obtain (3) by identifying $a \in A^m$ with $(a', a'') \in A^{m-1} \times A$.
2501.06662_F00170	9	■ 1.000	$i = m - 1$	$i = m - 1$	Then (4) follows by rewriting the term corresponding to $i = m - 1$ in the leftmost sum
2501.06662_F00171	9	■ 1.000	$\pi\left(x a^{\prime} \mid x\right) p(\dagger \mid x a^{\prime})$	$\pi(x a' \mid x) p(\dagger \mid x a')$	as $\pi(x a' \mid x) p(\dagger \mid x a')$, where $a' \in A^{m-1}$, and then moving it to the rightmost sum. The
2501.06662_F00172	9	■ 1.000	$a^{\prime} \in A^{m-1}$	$a' \in A^{m-1}$	as $\pi(x a' \mid x) p(\dagger \mid x a')$, where $a' \in A^{m-1}$, and then moving it to the rightmost sum. The
2501.06662_F00173	9	—	$\square$	$\square$	■
2501.06662_F00174	10	■ 0.561	$(\mathcal{V}, \leq, \otimes, 1)$	$(\mathcal{V}, \leq, \otimes, 1)$	2.11. DEFINITION. A <i>commutative monoidal preorder</i> $(\mathcal{V}, \leq, \otimes, 1)$ is a preordered set
2501.06662_F00175	10	■ 1.000	$(\mathcal{V}, \leq)$	$(\mathcal{V}, \leq)$	$(\mathcal{V}, \leq)$ and a commutative monoid $(\mathcal{V}, \otimes, 1)$ satisfying $x \otimes y \leq x' \otimes y'$ whenever $x \leq x'$
2501.06662_F00176	10	■ 1.000	$(\mathcal{V}, \otimes, 1)$	$(\mathcal{V}, \otimes, 1)$	$(\mathcal{V}, \leq)$ and a commutative monoid $(\mathcal{V}, \otimes, 1)$ satisfying $x \otimes y \leq x' \otimes y'$ whenever $x \leq x'$
2501.06662_F00177	10	■ 1.000	$x \otimes y \leq x^{\prime} \otimes y^{\prime}$	$x \otimes y \leq x' \otimes y'$	$(\mathcal{V}, \leq)$ and a commutative monoid $(\mathcal{V}, \otimes, 1)$ satisfying $x \otimes y \leq x' \otimes y'$ whenever $x \leq x'$
2501.06662_F00178	10	■ 1.000	$x \leq x^{\prime}$	$x \leq x'$	$(\mathcal{V}, \leq)$ and a commutative monoid $(\mathcal{V}, \otimes, 1)$ satisfying $x \otimes y \leq x' \otimes y'$ whenever $x \leq x'$
2501.06662_F00179	10	■ 1.000	$y \leq y^{\prime}$	$y \leq y'$	and $y \leq y'$.
2501.06662_F00180	10	■ 0.959	$([0, 1], \leq, \cdot, 1)$	$([0, 1], \leq, \cdot, 1)$	2.12. EXAMPLE. The unit interval $([0, 1], \leq, \cdot, 1)$ is a commutative monoidal preorder

Identifier	Page	Conf.	LaTeX source	Rendered	Image
2501.06662_F00181	10	0.554	$a \cdot b := a \cdot b$	$ab := a \cdot b$	we will denote by juxtaposition, $ab := a \cdot b$ for $a, b \in [0, 1]$, and the monoidal unit is 1.
2501.06662_F00182	10	0.554	$a, b \in [0, 1]$	$a, b \in [0, 1]$	we will denote by juxtaposition, $ab := a \cdot b$ for $a, b \in [0, 1]$, and the monoidal unit is 1.
2501.06662_F00183	10	0.842	$[0, \infty], \geq, +, 0$	$[0, \infty], \geq, +, 0$	2.13. EXAMPLE. The extended non-negative reals $([0, \infty], \geq, +, 0)$ form a commutative
2501.06662_F00184	10	1.000	$a + \infty := \infty$	$a + \infty := \infty$	The monoidal product is addition with $a + \infty := \infty$ and $\infty + a := \infty$ for all $a \in [0, \infty]$,
2501.06662_F00185	10	1.000	$\infty + a := \infty$	$\infty + a := \infty$	The monoidal product is addition with $a + \infty := \infty$ and $\infty + a := \infty$ for all $a \in [0, \infty]$,
2501.06662_F00186	10	1.000	$a \in [0, \infty]$	$a \in [0, \infty]$	The monoidal product is addition with $a + \infty := \infty$ and $\infty + a := \infty$ for all $a \in [0, \infty]$,
2501.06662_F00187	10	1.000	$\mathcal{V}$	$\mathcal{V}$	category enriched over $\mathcal{V}$, or simply a $\mathcal{V}$ -category, $\mathcal{C}$ consists of a set $\text{ob}(\mathcal{C})$ of objects and,
2501.06662_F00188	10	1.000	$\mathcal{C}$	$\mathcal{C}$	category enriched over $\mathcal{V}$, or simply a $\mathcal{V}$ -category, $\mathcal{C}$ consists of a set $\text{ob}(\mathcal{C})$ of objects and,
2501.06662_F00189	10	1.000	$\text{operatorname{ob}}(\mathcal{C})$	$\text{ob}(\mathcal{C})$	category enriched over $\mathcal{V}$, or simply a $\mathcal{V}$ -category, $\mathcal{C}$ consists of a set $\text{ob}(\mathcal{C})$ of objects and,
2501.06662_F00190	10	1.000	$\mathcal{C}(x, y)$	$\mathcal{C}(x, y)$	for every pair of objects x and y , an object $\mathcal{C}(x, y)$ of $\mathcal{V}$ called a $\mathcal{V}$ -hom object satisfying
2501.06662_F00191	10	0.998	$x, y, z \in \text{ob}(\mathcal{C})$	$x, y, z \in \text{ob}(\mathcal{C})$	the following: for all objects $x, y, z \in \text{ob}(\mathcal{C})$,
2501.06662_F00192	10	1.000	$\pi(y \mid x) = 0$	$\pi(y \mid x) = 0$	y from x one token at a time. Otherwise, if x is not contained in y , then $\pi(y \mid x) = 0$, and
2501.06662_F00193	10	1.000	$x = y$	$x = y$	if $x = y$ then $\pi(x \mid x) = 1$.
2501.06662_F00194	10	0.803	$\mathcal{L}(x, y) := \pi(y \mid x)$	$\mathcal{L}(x, y) := \pi(y \mid x)$	$[0, 1]$ -hom object to be $\mathcal{L}(x, y) := \pi(y \mid x)$.
2501.06662_F00195	11	0.939	x, y, z	x, y, z	a category enriched over $[0, 1]$ listed in Definition 2.14. Suppose first that x, y, z are three
2501.06662_F00196	11	1.000	$x \rightarrow y \rightarrow z$	$x \rightarrow y \rightarrow z$	strings satisfying $x \rightarrow y \rightarrow z$. If $x \neq y$ and $y \neq z$, then x and y are necessarily unfinished
2501.06662_F00197	11	1.000	$x \neq y$	$x \neq y$	strings satisfying $x \rightarrow y \rightarrow z$. If $x \neq y$ and $y \neq z$, then x and y are necessarily unfinished
2501.06662_F00198	11	1.000	$y \neq z$	$y \neq z$	strings satisfying $x \rightarrow y \rightarrow z$. If $x \neq y$ and $y \neq z$, then x and y are necessarily unfinished
2501.06662_F00199	11	1.000	$\left(a_i\right)_{i=1}^{t+k+k'-1} \subset A$	$(a_i)_{i=1}^{t+k+k'-1} \subset A$	for some $(a_i)_{i=1}^{t+k+k'-1} \subset A$ and $a_{t+k+k'} \in A \cup \{\dagger\}$. It then follows from (1) that
2501.06662_F00200	11	1.000	$a_{t+k+k'} \in A \cup \{\dagger\}$	$a_{t+k+k'} \in A \cup \{\dagger\}$	for some $(a_i)_{i=1}^{t+k+k'-1} \subset A$ and $a_{t+k+k'} \in A \cup \{\dagger\}$. It then follows from (1) that
2501.06662_F00201	11	1.000	$\pi(y \mid x) = 1$	$\pi(y \mid x) = 1$	If instead $x = y$, then $\pi(y \mid x) = 1$ and obviously $\pi(z \mid y) = \pi(z \mid x)$. The case $z = y$ is
2501.06662_F00202	11	1.000	$\pi(z \mid y) = \pi(z \mid x)$	$\pi(z \mid y) = \pi(z \mid x)$	If instead $x = y$, then $\pi(y \mid x) = 1$ and obviously $\pi(z \mid y) = \pi(z \mid x)$. The case $z = y$ is
2501.06662_F00203	11	1.000	$z = y$	$z = y$	If instead $x = y$, then $\pi(y \mid x) = 1$ and obviously $\pi(z \mid y) = \pi(z \mid x)$. The case $z = y$ is
2501.06662_F00204	11	0.593	z	z	treated similarly. More generally, if x, y and z are arbitrary objects of $\mathbf{L}$, one has that
2501.06662_F00205	11	0.873	$\pi(y \mid x) \pi(z \mid y) \leq \pi(z \mid x)$	$\pi(y \mid x) \pi(z \mid y) \leq \pi(z \mid x)$	$\pi(y \mid x) \pi(z \mid y) \leq \pi(z \mid x)$ as it may be the case that $\pi(y \mid x), \pi(z \mid x) \neq 0$ but $\pi(z \mid y) = 0$, ³
2501.06662_F00206	11	0.873	$\pi(y \mid x), \pi(z \mid x) \neq 0$	$\pi(y \mid x), \pi(z \mid x) \neq 0$	$\pi(y \mid x) \pi(z \mid y) \leq \pi(z \mid x)$ as it may be the case that $\pi(y \mid x), \pi(z \mid x) \neq 0$ but $\pi(z \mid y) = 0$, ³
2501.06662_F00207	11	0.873	$\pi(z \mid y) = 0, \{ \}^3$	$\pi(z \mid y) = 0,^3$	$\pi(y \mid x) \pi(z \mid y) \leq \pi(z \mid x)$ as it may be the case that $\pi(y \mid x), \pi(z \mid x) \neq 0$ but $\pi(z \mid y) = 0$, ³
2501.06662_F00208	11	1.000	$\pi(z \mid y), \pi(z \mid x) \neq 0$	$\pi(z \mid y), \pi(z \mid x) \neq 0$	or that $\pi(z \mid y), \pi(z \mid x) \neq 0$ but $\pi(y \mid x) = 0$. ⁴ In summary, we come to the following
2501.06662_F00209	11	1.000	$\pi(y \mid x) = 0, \{ \}^4$	$\pi(y \mid x) = 0.^4$	or that $\pi(z \mid y), \pi(z \mid x) \neq 0$ but $\pi(y \mid x) = 0$. ⁴ In summary, we come to the following
2501.06662_F00210	11	0.985	$[0, 1]$	$[0, 1]$	A more general version of this $[0, 1]$ -category was originally defined in [Bradley et al.,
2501.06662_F00211	11	1.000	$\mathcal{L}(x, y)$	$\mathcal{L}(x, y)$	2022, Definition 4], where $\mathcal{L}(x, y)$ was taken to be nonzero whenever x was an <i>arbitrary</i>
2501.06662_F00212	11	1.000	$x = \perp$	$x = \perp$	³ For example, consider when $x = \perp_{\text{green}}$ and $y = \perp_{\text{green}}$ lantern and $z = \perp_{\text{green}}$ salad.
2501.06662_F00213	11	1.000	$y = \perp$	$y = \perp$	³ For example, consider when $x = \perp_{\text{green}}$ and $y = \perp_{\text{green}}$ lantern and $z = \perp_{\text{green}}$ salad.
2501.06662_F00214	11	1.000	$z = \perp$	$z = \perp$	³ For example, consider when $x = \perp_{\text{green}}$ and $y = \perp_{\text{green}}$ lantern and $z = \perp_{\text{green}}$ salad.
2501.06662_F00215	12	1.000	$d(x, y) + d(y, z) \geq d(x, z)$	$d(x, y) + d(y, z) \geq d(x, z)$	It is straightforward to check the triangle inequality is satisfied $d(x, y) + d(y, z) \geq d(x, z)$
2501.06662_F00216	12	1.000	$d(x, x) = 0$	$d(x, x) = 0$	and further that $d(x, x) = 0$ for all strings x, y, z . (This follows from Equation (6) and the
2501.06662_F00217	12	0.444	∞	∞	this way, we obtain a category $\mathcal{M}$ enriched over $[0, \infty]$, also known as a <i>generalized metric</i>
2501.06662_F00218	12	1.000	$\mathcal{M}(x, y) := d(x, y)$	$\mathcal{M}(x, y) := d(x, y)$	strings x, y to be the distance $\mathcal{M}(x, y) := d(x, y)$. And now that we have a generalized
2501.06662_F00219	12	1.000	$\mathbf{L}^{\leq N}$	$\mathbf{L}^{\leq N}$	enriched category, which is an additional reason to consider the category $\mathbf{L}^{\leq N}$ with finite
2501.06662_F00220	12	1.000	R	R	[Meckes, 2017]. Briefly, one starts with a semi-ring R and a multiplicative function
2501.06662_F00221	12	1.000	$\ -\ : \text{ob}(\mathcal{V}) \rightarrow R$	$\ -\ : \text{ob}(\mathcal{V}) \rightarrow R$	$\ -\ : \text{ob}(\mathcal{V}) \rightarrow R$ called <i>size</i> that is invariant under isomorphisms. So, $\ v\ = \ w\ $
2501.06662_F00222	12	1.000	$\ v\ = \ w\ $	$\ v\ = \ w\ $	$\ -\ : \text{ob}(\mathcal{V}) \rightarrow R$ called <i>size</i> that is invariant under isomorphisms. So, $\ v\ = \ w\ $
2501.06662_F00223	12	0.981	$v \cong w$	$v \cong w$	whenever $v \cong w$ in $\mathcal{V}$, and $\ 1\ = 1$ and $\ v \otimes w\ = \ v\ \ w\ $ for all $v, w \in \text{ob}(\mathcal{V})$.
2501.06662_F00224	12	0.981	$\ 1\ = 1$	$\ 1\ = 1$	whenever $v \cong w$ in $\mathcal{V}$, and $\ 1\ = 1$ and $\ v \otimes w\ = \ v\ \ w\ $ for all $v, w \in \text{ob}(\mathcal{V})$.
2501.06662_F00225	12	0.981	$\ v \otimes w\ = \ v\ \ w\ $	$\ v \otimes w\ = \ v\ \ w\ $	whenever $v \cong w$ in $\mathcal{V}$, and $\ 1\ = 1$ and $\ v \otimes w\ = \ v\ \ w\ $ for all $v, w \in \text{ob}(\mathcal{V})$.
2501.06662_F00226	12	0.981	$v, w \in \text{ob}(\mathcal{V})$	$v, w \in \text{ob}(\mathcal{V})$	whenever $v \cong w$ in $\mathcal{V}$, and $\ 1\ = 1$ and $\ v \otimes w\ = \ v\ \ w\ $ for all $v, w \in \text{ob}(\mathcal{V})$.
2501.06662_F00227	12	1.000	$\zeta_{\mathcal{C}}: \text{ob}(\mathcal{C}) \times \text{ob}(\mathcal{C}) \rightarrow R$	$\zeta_{\mathcal{C}}: \text{ob}(\mathcal{C}) \times \text{ob}(\mathcal{C}) \rightarrow R$	$\zeta_{\mathcal{C}}: \text{ob}(\mathcal{C}) \times \text{ob}(\mathcal{C}) \rightarrow R$, defined by $\zeta_{\mathcal{C}}(x, y) := \ \mathcal{C}(x, y)\ $. The function $\zeta_{\mathcal{C}}$ can be regarded
2501.06662_F00228	12	1.000	$\zeta_{\mathcal{C}}(x, y) := \ \mathcal{C}(x, y)\ $	$\zeta_{\mathcal{C}}(x, y) := \ \mathcal{C}(x, y)\ $	$\zeta_{\mathcal{C}}: \text{ob}(\mathcal{C}) \times \text{ob}(\mathcal{C}) \rightarrow R$, defined by $\zeta_{\mathcal{C}}(x, y) := \ \mathcal{C}(x, y)\ $. The function $\zeta_{\mathcal{C}}$ can be regarded
2501.06662_F00229	12	1.000	$\zeta_{\mathcal{C}}$	$\zeta_{\mathcal{C}}$	$\zeta_{\mathcal{C}}: \text{ob}(\mathcal{C}) \times \text{ob}(\mathcal{C}) \rightarrow R$, defined by $\zeta_{\mathcal{C}}(x, y) := \ \mathcal{C}(x, y)\ $. The function $\zeta_{\mathcal{C}}$ can be regarded
2501.06662_F00230	12	1.000	$\mu_{\mathcal{C}} := \zeta_{\mathcal{C}}^{-1}$	$\mu_{\mathcal{C}} := \zeta_{\mathcal{C}}^{-1}$	to have <i>Möbius inversion</i> and its <i>Möbius function</i> is $\mu_{\mathcal{C}} := \zeta_{\mathcal{C}}^{-1}$ (the entries of this matrix
2501.06662_F00231	12	1.000	$\text{Mag}(\mathcal{C})$	$\text{Mag}(\mathcal{C})$	are called <i>Möbius coefficients</i>). When $\mathcal{C}$ has Möbius inversion, the <i>magnitude</i> $\text{Mag}(\mathcal{C})$ of

Identifier	Page	Conf.	LaTeX source	Rendered	Image
2501.06662_F00232	13	■ 1.000	$\mathcal{V} = [\emptyset, \infty]$	$\mathcal{V} = [0, \infty]$	For $\mathcal{V} = [0, \infty]$, every size function is of the form $\ x\ _t = e^{-tx}$ for some $t \in \mathbb{R} \cup \{\infty\}$. It
2501.06662_F00233	13	■ 1.000	$ x _t = e^{-tx}$	$\ x\ _t = e^{-tx}$	For $\mathcal{V} = [0, \infty]$, every size function is of the form $\ x\ _t = e^{-tx}$ for some $t \in \mathbb{R} \cup \{\infty\}$. It
2501.06662_F00234	13	■ 1.000	$t \in \mathbb{R} \cup \{\infty\}$	$t \in \mathbb{R} \cup \{\infty\}$	For $\mathcal{V} = [0, \infty]$, every size function is of the form $\ x\ _t = e^{-tx}$ for some $t \in \mathbb{R} \cup \{\infty\}$. It
2501.06662_F00235	13	■ 1.000	$\ -\ _1$	$\ -\ _1$	is customary to choose $\ -\ _1$ and then consider the <i>magnitude function</i> $f(t) := \text{Mag}(t\mathcal{C})$,
2501.06662_F00236	13	■ 1.000	$f(t) := \text{operatorname{Mag}}(t \mathcal{C})$	$f(t) := \text{Mag}(t\mathcal{C})$	is customary to choose $\ -\ _1$ and then consider the <i>magnitude function</i> $f(t) := \text{Mag}(t\mathcal{C})$,
2501.06662_F00237	13	■ 1.000	$t \in (\emptyset, \infty)$	$t \in (0, \infty)$	defined for $t \in (0, \infty)$ [Leinster, 2013, Section 2.2]. The symbol $t\mathcal{C}$ denotes the generalized
2501.06662_F00238	13	■ 1.000	$t \mathcal{C}$	$t\mathcal{C}$	defined for $t \in (0, \infty)$ [Leinster, 2013, Section 2.2]. The symbol $t\mathcal{C}$ denotes the generalized
2501.06662_F00239	13	■ 1.000	$[s, t] = \{u \in P : s \leq u \leq t\}$	$[s, t] = \{u \in P : s \leq u \leq t\}$	Given a poset P , define a <i>closed interval</i> to be $[s, t] = \{u \in P : s \leq u \leq t\}$ whenever
2501.06662_F00240	13	■ 1.000	$s \leq t$	$s \leq t$	$s \leq t$ in P . If every interval of P is finite, then P is said to be a <i>locally finite</i> poset.
2501.06662_F00241	13	■ 1.000	$\mathbb{k}$	$\mathbb{k}$	Let $\mathbb{k}$ be a field, P a locally finite poset, and $\text{int}(P)$ the set of all closed intervals of P .
2501.06662_F00242	13	■ 1.000	$\text{operatorname{int}}(P)$	$\text{int}(P)$	Let $\mathbb{k}$ be a field, P a locally finite poset, and $\text{int}(P)$ the set of all closed intervals of P .
2501.06662_F00243	13	■ 1.000	$I(P, \mathbb{k})$	$I(P, \mathbb{k})$	The <i>incidence algebra</i> $I(P, \mathbb{k})$ of P over $\mathbb{k}$ is the $\mathbb{k}$ -algebra of all functions $f : \text{int}(P) \rightarrow \mathbb{k}$
2501.06662_F00244	13	■ 1.000	$f : \text{operatorname{int}}(P) \rightarrow \mathbb{k}$	$f : \text{int}(P) \rightarrow \mathbb{k}$	The <i>incidence algebra</i> $I(P, \mathbb{k})$ of P over $\mathbb{k}$ is the $\mathbb{k}$ -algebra of all functions $f : \text{int}(P) \rightarrow \mathbb{k}$
2501.06662_F00245	13	■ 0.978	$f(s, t)$	$f(s, t)$	where we write $f(s, t)$ for $f([s, t])$. (Remark that the sum has finitely many terms.) The
2501.06662_F00246	13	■ 0.978	$f([s, t])$	$f([s, t])$	where we write $f(s, t)$ for $f([s, t])$. (Remark that the sum has finitely many terms.) The
2501.06662_F00247	13	■ 0.996	$\delta : \text{operatorname{int}}(P) \rightarrow \mathbb{k}$	$\delta : \text{int}(P) \rightarrow \mathbb{k}$	incidence algebra of P is an associative algebra with identity $\delta : \text{int}(P) \rightarrow \mathbb{k}$ given by
2501.06662_F00248	13	■ 1.000	$f * \delta = \delta * f = f$	$f * \delta = \delta * f = f$	since $f * \delta = \delta * f = f$ for all functions $f \in I(P, \mathbb{k})$. The <i>zeta function</i> ζ_P is another
2501.06662_F00249	13	■ 1.000	$f \in I(P, \mathbb{k})$	$f \in I(P, \mathbb{k})$	since $f * \delta = \delta * f = f$ for all functions $f \in I(P, \mathbb{k})$. The <i>zeta function</i> ζ_P is another
2501.06662_F00250	13	■ 1.000	ζ_P	ζ_P	since $f * \delta = \delta * f = f$ for all functions $f \in I(P, \mathbb{k})$. The <i>zeta function</i> ζ_P is another
2501.06662_F00251	13	■ 0.754	$\mathbf{2} := \{\text{true}, \text{false}\}$	$\mathbf{2} := \{\text{true}, \text{false}\}$	category enriched over truth values $\mathbf{2} := \{\text{true}, \text{false}\}$. Its objects are the elements of
2501.06662_F00252	13	■ 0.754	$\mathbb{2}$	$\mathbb{2}$	category enriched over truth values $\mathbf{2} := \{\text{true}, \text{false}\}$. Its objects are the elements of
2501.06662_F00253	13	■ 1.000	$s, u \in P$	$s, u \in P$	P , and given $s, u \in P$, if $s \leq u$, then their corresponding hom object $P(s, u)$ is true , and
2501.06662_F00254	13	■ 1.000	$s \leq u$	$s \leq u$	P , and given $s, u \in P$, if $s \leq u$, then their corresponding hom object $P(s, u)$ is true , and
2501.06662_F00255	13	■ 1.000	$P(s, u)$	$P(s, u)$	P , and given $s, u \in P$, if $s \leq u$, then their corresponding hom object $P(s, u)$ is true , and
2501.06662_F00256	13	■ 1.000	$s \not\leq u$	$s \not\leq u$	if $s \not\leq u$ then it is false . The identity and compositionality requirements in Definition
2501.06662_F00257	13	■ 0.999	$\ -\ : \text{operatorname{ob}}(\mathbf{2}) \rightarrow \mathbb{Z}$	$\ -\ : \text{ob}(\mathbf{2}) \rightarrow \mathbb{Z}$	size function $\ -\ : \text{ob}(\mathbf{2}) \rightarrow \mathbb{Z}$ given by $\ \text{true}\ = 1$ and $\ \text{false}\ = 0$ and then setting
2501.06662_F00258	13	■ 0.999	$\ -\ $	$\ -\ $	size function $\ -\ : \text{ob}(\mathbf{2}) \rightarrow \mathbb{Z}$ given by $\ \text{true}\ = 1$ and $\ \text{false}\ = 0$ and then setting
2501.06662_F00259	13	■ 0.999	$\ -\ = 1$	$\ -\ = 1$	size function $\ -\ : \text{ob}(\mathbf{2}) \rightarrow \mathbb{Z}$ given by $\ \text{true}\ = 1$ and $\ \text{false}\ = 0$ and then setting
2501.06662_F00260	13	■ 0.999	$\ -\ = 0$	$\ -\ = 0$	size function $\ -\ : \text{ob}(\mathbf{2}) \rightarrow \mathbb{Z}$ given by $\ \text{true}\ = 1$ and $\ \text{false}\ = 0$ and then setting
2501.06662_F00261	13	■ 1.000	$\zeta_P(s, u) = \ P(s, u)\ $	$\zeta_P(s, u) = \ P(s, u)\ $	$\zeta_P(s, u) = \ P(s, u)\ $.
2501.06662_F00262	13	■ 1.000	μ_P	μ_P	Subsection 3.1. The inverse of ζ_P is called the <i>Möbius function</i> of P , denoted μ_P , and it
2501.06662_F00263	14	■ 0.576	$\mu_{\mathbf{L}}$	$\mu_{\mathbf{L}}$	3.3. EXAMPLE. The Möbius function $\mu_{\mathbf{L}}$ on the poset $\mathbf{L}$ of strings is rather simple. This is
2501.06662_F00264	14	■ 0.997	$x \in \text{operatorname{ob}}(\mathbf{L})$	$x \in \text{ob}(\mathbf{L})$	$x \in \text{ob}(\mathbf{L})$ and tokens $a_1, a_2, \dots \in A \cup \{\dagger\}$, we have
2501.06662_F00265	14	■ 0.997	$a_1, a_2, \dots \in A \cup \{\dagger\}$	$a_1, a_2, \dots \in A \cup \{\dagger\}$	$x \in \text{ob}(\mathbf{L})$ and tokens $a_1, a_2, \dots \in A \cup \{\dagger\}$, we have
2501.06662_F00266	14	■ 1.000	$\mu_{\mathbf{L}}(x, xa_1a_2 \cdots a_j) = 0$	$\mu_{\mathbf{L}}(x, xa_1a_2 \cdots a_j) = 0$	Similarly, $\mu_{\mathbf{L}}(x, xa_1a_2 \cdots a_j) = 0$ for $j > 1$. (As one of the reviewers pointed out, these
2501.06662_F00267	14	■ 1.000	$j > 1$	$j > 1$	Similarly, $\mu_{\mathbf{L}}(x, xa_1a_2 \cdots a_j) = 0$ for $j > 1$. (As one of the reviewers pointed out, these
2501.06662_F00268	14	■ 0.677	$\#A$	$\#A$	let $\#A$ be the cardinality of the token set A and let $\#\text{ob}(\mathbf{L})$ be the number of strings in
2501.06662_F00269	14	■ 0.677	$\#$	$\#$	let $\#A$ be the cardinality of the token set A and let $\#\text{ob}(\mathbf{L})$ be the number of strings in
2501.06662_F00270	14	■ 0.677	$(\mathbf{L})$	$(\mathbf{L})$	let $\#A$ be the cardinality of the token set A and let $\#\text{ob}(\mathbf{L})$ be the number of strings in
2501.06662_F00271	15	■ 0.585	$\mathbf{L}^{\leq N} = \bigsqcup_{i=0}^{N-1} \mathbf{L}^{(i)}$	$\mathbf{L}^{\leq N} = \bigsqcup_{i=0}^{N-1} \mathbf{L}^{(i)}$	Unwinding the last line, recall that $\mathbf{L}$ denotes $\mathbf{L}^{\leq N} = \bigsqcup_{i=0}^{N-1} \mathbf{L}^{(i)}$. We have $\mathbf{L}^{(0)} = \{\perp\}$
2501.06662_F00272	15	■ 0.585	$\mathbf{L}^{(0)} = \{\perp\}$	$\mathbf{L}^{(0)} = \{\perp\}$	Unwinding the last line, recall that $\mathbf{L}$ denotes $\mathbf{L}^{\leq N} = \bigsqcup_{i=0}^{N-1} \mathbf{L}^{(i)}$. We have $\mathbf{L}^{(0)} = \{\perp\}$
2501.06662_F00273	15	■ 0.508	$i \leq 1, \mathbf{L}^{(i)}$	$i \leq 1, \mathbf{L}^{(i)}$	and, for each $i \leq 1$, $\mathbf{L}^{(i)}$ is comprised of strings of the form $\perp a$ with $a \in A^i$ or $\perp a'\dagger$ with
2501.06662_F00274	15	■ 0.508	$a \in A^i$	$a \in A^i$	and, for each $i \leq 1$, $\mathbf{L}^{(i)}$ is comprised of strings of the form $\perp a$ with $a \in A^i$ or $\perp a'\dagger$ with
2501.06662_F00275	15	■ 0.508	$\perp a'\dagger$	$\perp a'\dagger$	and, for each $i \leq 1$, $\mathbf{L}^{(i)}$ is comprised of strings of the form $\perp a$ with $a \in A^i$ or $\perp a'\dagger$ with
2501.06662_F00276	15	■ 1.000	$a' \in A^{i-1}$	$a' \in A^{i-1}$	$a' \in A^{i-1}$. It follows that $\#\mathbf{L}^{(i)} = \#A^i + \#A^{i-1}$ and so
2501.06662_F00277	15	■ 1.000	$\# \mathbf{L}^{(i)} = \#A^i + \#A^{i-1}$	$\# \mathbf{L}^{(i)} = \#A^i + \#A^{i-1}$	$a' \in A^{i-1}$. It follows that $\#\mathbf{L}^{(i)} = \#A^i + \#A^{i-1}$ and so
2501.06662_F00278	15	■ 0.998	$\text{operatorname{Mag}}(\mathbf{L}) = 1$	$\text{Mag}(\mathbf{L}) = 1$	Therefore, $\text{Mag}(\mathbf{L}) = 1$.
2501.06662_F00279	15	■ 0.999	$\text{operatorname{ob}}(\mathcal{M}) = \text{operatorname{ob}}(\mathbf{L}^{\leq N})$	$\text{ob}(\mathcal{M}) = \text{ob}(\mathbf{L}^{\leq N})$	introduced in Subsection 2.17, whose objects are $\text{ob}(\mathcal{M}) = \text{ob}(\mathbf{L}^{\leq N})$ and where the $[0, \infty]$ -

Identifier	Page	Conf.	LaTeX source	Rendered	Image
2501.06662_F00280	15	■ 1.000	$\mathcal{M}(x, y) = d(x, y) = -\ln \pi(y \mid x)$	$\mathcal{M}(x, y) = d(x, y) = -\ln \pi(y \mid x)$	hom objects are given by $\mathcal{M}(x, y) = d(x, y) = -\ln \pi(y x)$. Define the <i>zeta matrix</i>
2501.06662_F00281	15	■ 1.000	$\zeta_{\mathcal{M}} : \text{ob}(\mathcal{M}) \times \text{ob}(\mathcal{M}) \rightarrow \mathbb{R}$	$\zeta_{\mathcal{M}} : \text{ob}(\mathcal{M}) \times \text{ob}(\mathcal{M}) \rightarrow \mathbb{R}$	$\zeta_{\mathcal{M}} : \text{ob}(\mathcal{M}) \times \text{ob}(\mathcal{M}) \rightarrow \mathbb{R}$ to be
2501.06662_F00282	15	■ 1.000	$d(x, y) = \infty$	$d(x, y) = \infty$	It is understood that $d(x, y) = \infty$ whenever $\pi(y x) = 0$; for instance, the distance from
2501.06662_F00283	15	■ 1.000	ζ_t	ζ_t	a finished text x to any other string y is infinite. For simplicity, we will write ζ_t instead
2501.06662_F00284	15	■ 1.000	$\left(\zeta_{\mathcal{M}}\right)_t$	$(\zeta_{\mathcal{M}})_t$	of $(\zeta_{\mathcal{M}})_t$. As we will see in the proposition and corollaries below, ζ_t^{-1} exists and its
2501.06662_F00285	15	■ 1.000	ζ_t^{-1}	ζ_t^{-1}	of $(\zeta_{\mathcal{M}})_t$. As we will see in the proposition and corollaries below, ζ_t^{-1} exists and its
2501.06662_F00286	15	■ 0.988	$\text{ob}(\mathcal{M})$	$\text{ob}(\mathcal{M})$	$\text{ob}(\mathcal{M})$,
2501.06662_F00287	15	■ 1.000	$x=y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_k=y$	$x = y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_k = y$	(A <i>non-degenerate path</i> $\overline{x=y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_k=y}$ consists of a sequence of composable
2501.06662_F00288	16	■ 1.000	δ	δ	where δ is the identity matrix: $\delta(x, y) = 1$ if $x = y$ and $\delta(x, y) = 0$ otherwise.
2501.06662_F00289	16	■ 1.000	$\delta(x, y) = 1$	$\delta(x, y) = 1$	where δ is the identity matrix: $\delta(x, y) = 1$ if $x = y$ and $\delta(x, y) = 0$ otherwise.
2501.06662_F00290	16	■ 1.000	$\delta(x, y) = 0$	$\delta(x, y) = 0$	where δ is the identity matrix: $\delta(x, y) = 1$ if $x = y$ and $\delta(x, y) = 0$ otherwise.
2501.06662_F00291	16	■ 0.997	$\zeta - \delta$	$\zeta - \delta$	The matrix $\zeta - \delta$ is the adjacency matrix of a directed graph D with vertices $\text{ob}(\mathcal{M})$
2501.06662_F00292	16	■ 0.997	D	D	The matrix $\zeta - \delta$ is the adjacency matrix of a directed graph D with vertices $\text{ob}(\mathcal{M})$
2501.06662_F00293	16	■ 0.412	$E = \{(z, z') \mid z \rightarrow z'\}$	$E = \{(z, z') \mid z \rightarrow z'\}$	(equivalently, $\text{ob}(\mathbf{L})$) and edges $\overline{E = \{(z, z') \mid z \rightarrow z'\}}$ in $\mathbf{L}$, $z \neq z'$) (we are excluding edges
2501.06662_F00294	16	■ 0.412	$\mathbf{L}, z \neq z'$	$\mathbf{L}, z \neq z'$	(equivalently, $\text{ob}(\mathbf{L})$) and edges $E = \{(z, z') \mid z \rightarrow z' \text{ in } \mathbf{L}, z \neq z'\}$ (we are excluding edges
2501.06662_F00295	16	■ 1.000	$(\zeta_t - \delta)^k$	$(\zeta_t - \delta)^k$	coming from identity morphisms, whose weights vanish). The power $(\zeta_t - \delta)^k$ counts non-
2501.06662_F00296	16	■ 0.672	k	k	degenerate paths of length k in D . Since $\mathbf{L}$ is a finite poset, one readily verifies that E
2501.06662_F00297	16	■ 0.672	E	E	degenerate paths of length k in D . Since $\mathbf{L}$ is a finite poset, one readily verifies that E
2501.06662_F00298	16	■ 0.990	(x, y)	(x, y)	The expression (8) arises from evaluating (9) on (x, y) . In fact,
2501.06662_F00299	16	■ 1.000	$\zeta_{\mathbf{L}}^{-1}$	$\zeta_{\mathbf{L}}^{-1}$	a simplified expression in terms of π from Definition 2.8 and the Möbius function $\zeta_{\mathbf{L}}^{-1}$,
2501.06662_F00300	16	■ 1.000	$\zeta_{\mathbf{L}}$	$\zeta_{\mathbf{L}}$	whose zeta function $\zeta_{\mathbf{L}}$ was described in Section 3.2.
2501.06662_F00301	17	■ 0.998	$k \geq 0$	$k \geq 0$	PROOF. For any $k \geq 0$ and any nondegenerate path $x = y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_k = y$ in $\mathbf{L}$,
2501.06662_F00302	17	■ 1.000	$\zeta_t^{-1}(x, y)$	$\zeta_t^{-1}(x, y)$	Hence the Möbius coefficients $\zeta_t^{-1}(x, y)$ can be written as
2501.06662_F00303	17	■ 0.999	$\pi(y \mid x)^t \zeta_{\mathbf{L}}^{-1}(x, y)$	$\pi(y \mid x)^t \zeta_{\mathbf{L}}^{-1}(x, y)$	which is equal to $\pi(y x)^t \zeta_{\mathbf{L}}^{-1}(x, y)$ by Philip Hall’s theorem [Hall, 1936], cf. [Stanley, 2011,
2501.06662_F00304	17	■ 1.000	$\zeta_{\mathbf{L}}^{-1}(x, y) = \mu_{\mathbf{L}}(x, y)$	$\zeta_{\mathbf{L}}^{-1}(x, y) = \mu_{\mathbf{L}}(x, y)$	Now recall that $\zeta_{\mathbf{L}}^{-1}(x, y) = \mu_{\mathbf{L}}(x, y)$, and so ζ_t^{-1} can be simplified even further by
2501.06662_F00305	17	■ 0.988	$f : (0, \infty) \rightarrow \mathbb{R}$	$f : (0, \infty) \rightarrow \mathbb{R}$	generalized metric space $\mathcal{M}$. By Equation (7) it is the function $\overline{f : (0, \infty) \rightarrow \mathbb{R}}$ given by
2501.06662_F00306	17	■ 0.907	$f(t) = \text{Mag}(t\mathcal{M})$	$f(t) = \text{Mag}(t\mathcal{M})$	$\overline{f(t) = \text{Mag}(t\mathcal{M})}$, where ⁶
2501.06662_F00307	17	■ 1.000	$p = (p_1, \dots, p_n)$	$p = (p_1, \dots, p_n)$	tion $\overline{p = (p_1, \dots, p_n)}$ is equal to
2501.06662_F00308	17	■ 0.885	$(p_x : A \cup \{\dagger\} \rightarrow [0, 1])_{x \in \text{ob}(\mathbf{L})}$	$(p_x : A \cup \{\dagger\} \rightarrow [0, 1])_{x \in \text{ob}(\mathbf{L})}$	rithmic entropies of the probability distributions $(p_x : A \cup \{\dagger\} \rightarrow [0, 1])_{x \in \text{ob}(\mathbf{L})}$ generated
2501.06662_F00309	17	■ 1.000	$T(\perp)$	$T(\perp)$	by a language model and the cardinality of the terminal states $T(\perp)$ of the beginning-
2501.06662_F00310	17	■ 0.998	$\text{Mag}(t\mathcal{M}) = \sum_{x, y \in \text{ob}(\mathcal{M})} \pi(y \mid x)^t \zeta_{\mathbf{L}}^{-1}(x, y)$	$\text{Mag}(t\mathcal{M}) = \sum_{x, y \in \text{ob}(\mathcal{M})} \pi(y \mid x)^t \zeta_{\mathbf{L}}^{-1}(x, y)$	⁶ By Corollary 3.8, it follows that $\overline{\text{Mag}(t\mathcal{M}) = \sum_{x, y \in \text{ob}(\mathcal{M})} \pi(y x)^t \zeta_{\mathbf{L}}^{-1}(x, y)}$, which is a weighted version
2501.06662_F00311	18	■ 1.000	$y \neq x$	$y \neq x$	strings x that are not the prefix of any other string $y \neq x$, such as all finished texts. The
2501.06662_F00312	18	■ 1.000	$\text{ob}(\mathcal{M}) \setminus T(\perp)$	$\text{ob}(\mathcal{M}) \setminus T(\perp)$	single token, which is why the indexing set is $\text{ob}(\mathcal{M}) \setminus T(\perp)$. Equation (11) follows from
2501.06662_F00313	18	■ 0.996	$\text{ob}(\mathcal{M}) = T(\perp) \sqcup (\text{ob}(\mathcal{M}) \setminus T(\perp))$	$\text{ob}(\mathcal{M}) = T(\perp) \sqcup (\text{ob}(\mathcal{M}) \setminus T(\perp))$	the decomposition $\overline{\text{ob}(\mathcal{M}) = T(\perp) \sqcup (\text{ob}(\mathcal{M}) \setminus T(\perp))}$ and the fact that $\zeta_t^{-1}(x, x) = 1$ for
2501.06662_F00314	18	■ 0.996	$\zeta_t^{-1}(x, x) = 1$	$\zeta_t^{-1}(x, x) = 1$	the decomposition $\text{ob}(\mathcal{M}) = T(\perp) \sqcup (\text{ob}(\mathcal{M}) \setminus T(\perp))$ and the fact that $\overline{\zeta_t^{-1}(x, x) = 1}$ for
2501.06662_F00315	18	■ 0.998	$x \in \text{ob}(\mathcal{M})$	$x \in \text{ob}(\mathcal{M})$	all $x \in \text{ob}(\mathcal{M})$.
2501.06662_F00316	18	■ 1.000	$t \neq 1$	$t \neq 1$	on $A \cup \{\dagger\}$. When $t \neq 1$, we can multiply the sum by $(t-1)/(t-1)$ and rearrange factors
2501.06662_F00317	18	■ 1.000	$(t-1)/(t-1)$	$(t-1)/(t-1)$	on $A \cup \{\dagger\}$. When $t \neq 1$, we can multiply the sum by $\overline{(t-1)/(t-1)}$ and rearrange factors
2501.06662_F00318	18	■ 1.000	$t \geq 1$	$t \geq 1$	of the effective size of a language model’s linguistic space, in the sense that for $t \geq 1$,
2501.06662_F00319	18	■ 1.000	$p : S \rightarrow [0, 1]$	$p : S \rightarrow [0, 1]$	3.11. REMARK. It is easy to show (e.g. via L’Hôpital’s rule) that if $\overline{p : S \rightarrow [0, 1]}$ is a
2501.06662_F00320	18	■ 1.000	S	S	probability mass function on a finite set S ,
2501.06662_F00321	18	■ 1.000	$H(p)$	$H(p)$	The quantity $H(p)$ is the <i>Shannon entropy</i> of p (in nats). This entails that Shannon en-
2501.06662_F00322	18	■ 1.000	p	p	The quantity $H(p)$ is the <i>Shannon entropy</i> of p (in nats). This entails that Shannon en-

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2501.06662_F00323	19	■ 1.000	$E_{\{x\}}(a)=$	$E_x(a) =$	is the <i>partition function</i> of a system with microstates $A \cup \{\dagger\}$ and internal energy $E_x(a) =$
2501.06662_F00324	19	■ 1.000	$-\ln p_{\{x\}}(a)=d(x, x \ a)$	$-\ln p_x(a) = d(x, xa)$	$-\ln p_x(a) = d(x, xa)$ at inverse temperature $t > 0$. One can verify (see, for instance, [Baez,
2501.06662_F00325	19	■ 1.000	$f(t)=$	$f(t) =$	Since internal energy does not have to be normalized, we could also write $f(t) =$
2501.06662_F00326	19	■ 1.000	$\#(\operatorname{ob}(\mathcal{M}))-\tilde{Z}(t)$	$\#(\operatorname{ob}(\mathcal{M})) - \tilde{Z}(t)$	$\#(\operatorname{ob}(\mathcal{M})) - \tilde{Z}(t)$, where $\tilde{Z}$ is the partition function of a system with microstates $(x, a) \in$
2501.06662_F00327	19	■ 1.000	$\tilde{Z}$	$\tilde{Z}$	$\#(\operatorname{ob}(\mathcal{M})) - \tilde{Z}(t)$, where $\tilde{Z}$ is the partition function of a system with microstates $(x, a) \in$
2501.06662_F00328	19	■ 1.000	$(x, a) \in$	$(x, a) \in$	$\#(\operatorname{ob}(\mathcal{M})) - \tilde{Z}(t)$, where $\tilde{Z}$ is the partition function of a system with microstates $(x, a) \in$
2501.06662_F00329	19	■ 0.884	$S:=(\operatorname{ob}(\mathcal{M}) \backslash T(\perp)) \times (A \cup \{\dagger\})$ $\times (A \cup \{\dagger\})$	$S := (\operatorname{ob}(\mathcal{M}) \backslash T(\perp)) \times (A \cup \{\dagger\})$	$S := (\operatorname{ob}(\mathcal{M}) \backslash T(\perp)) \times (A \cup \{\dagger\})$ and internal energy $E(x, a) = d(x, a) = -\ln p_x(a)$.
2501.06662_F00330	19	■ 0.884	$E(x, a)=d(x, a)=-\ln p_{\{x\}}(a)$	$E(x, a) = d(x, a) = -\ln p_x(a)$	$S := (\operatorname{ob}(\mathcal{M}) \backslash T(\perp)) \times (A \cup \{\dagger\})$ and internal energy $E(x, a) = d(x, a) = -\ln p_x(a)$.
2501.06662_F00331	19	■ 1.000	ρ	ρ	there is a corresponding probability mass function (Gibbs state) ρ on S , given by $\rho(s) =$
2501.06662_F00332	19	■ 1.000	$\rho(s)=$	$\rho(s) =$	there is a corresponding probability mass function (Gibbs state) ρ on S , given by $\rho(s) =$
2501.06662_F00333	19	■ 1.000	$e^{-t E(s)} / \tilde{Z}(t)$	$e^{-t E(s)} / \tilde{Z}(t)$	$e^{-t E(s)} / \tilde{Z}(t)$. Then
2501.06662_F00334	19	■ 1.000	$y, z \in \operatorname{ob}(\mathcal{M}_{\{x\}})$	$y, z \in \operatorname{ob}(\mathcal{M}_x)$	the hom-object between any $y, z \in \operatorname{ob}(\mathcal{M}_x)$ to be $\mathcal{M}_x(y, z) := \mathcal{M}(y, z) = -\ln \pi(z y)$.
2501.06662_F00335	19	■ 1.000	$\mathcal{M}_{\{x\}}(y, z):=\mathcal{M}(y, z)=-\ln \pi(z \mid y)$	$\mathcal{M}_x(y, z) := \mathcal{M}(y, z) = -\ln \pi(z \mid y)$	the hom-object between any $y, z \in \operatorname{ob}(\mathcal{M}_x)$ to be $\mathcal{M}_x(y, z) := \mathcal{M}(y, z) = -\ln \pi(z y)$.
2501.06662_F00336	20	■ 0.644	M, d	M, d	magnitude function of a metric space (M, d) as a weighted sum of Euler characteristics
2501.06662_F00337	20	■ 1.000	$\ell \in [0, \infty)$	$\ell \in [0, \infty)$	of certain chain complexes. More precisely, for each $\ell \in [0, \infty)$, there is a chain complex
2501.06662_F00338	20	■ 1.000	$\left(MC_{\{k, \ell\}}(M)\right)_{k \in \mathbb{N}}$	$(MC_{k, \ell}(M))_{k \in \mathbb{N}}$	comprised of free abelian groups $(MC_{k, \ell}(M))_{k \in \mathbb{N}}$ such that $MC_{k, \ell}(M) := \mathbb{Z}[G_{k, \ell}]$ where
2501.06662_F00339	20	■ 1.000	$MC_{\{k, \ell\}}(M):=\mathbb{Z}[G_{\{k, \ell\}}]$	$MC_{k, \ell}(M) := \mathbb{Z}[G_{k, \ell}]$	comprised of free abelian groups $(MC_{k, \ell}(M))_{k \in \mathbb{N}}$ such that $MC_{k, \ell}(M) := \mathbb{Z}[G_{k, \ell}]$ where
2501.06662_F00340	20	■ 1.000	$\partial_k: MC_{\{k, \ell\}}(M) \rightarrow MC_{\{k-1, \ell\}}(M)$	$\partial_k : MC_{k, \ell}(M) \rightarrow MC_{k-1, \ell}(M)$	The differential $\partial_k : MC_{k, \ell}(M) \rightarrow MC_{k-1, \ell}(M)$ is defined as an alternating sum
2501.06662_F00341	20	■ 1.000	$0 < i < k$	$0 < i < k$	where for each $0 < i < k$,
2501.06662_F00342	20	■ 0.704	$\partial_k^0 = \partial_k^k = 0$	$\partial_k^0 = \partial_k^k = 0$	and $\partial_k^0 = \partial_k^k = 0$. The resulting $[0, \infty)$ -graded chain complex $(MC_{\bullet, \ell}(M), \partial_{\bullet})$ is called the
2501.06662_F00343	20	■ 0.704	$[0, \infty)$	$[0, \infty)$	and $\partial_k^0 = \partial_k^k = 0$. The resulting $[0, \infty)$ -graded chain complex $(MC_{\bullet, \ell}(M), \partial_{\bullet})$ is called the
2501.06662_F00344	20	■ 0.704	$\left(MC_{\{\bullet, \ell\}}(M), \partial_{\bullet}\right)$	$(MC_{\bullet, \ell}(M), \partial_{\bullet})$	and $\partial_k^0 = \partial_k^k = 0$. The resulting $[0, \infty)$ -graded chain complex $(MC_{\bullet, \ell}(M), \partial_{\bullet})$ is called the
2501.06662_F00345	20	■ 1.000	M	M	<i>magnitude complex</i> of M [Leinster and Shulman, 2021, Definition 3.3] and its homology
2501.06662_F00346	20	■ 0.892	$H_{\{\bullet, \ell\}}(M)$	$H_{\bullet, \ell}(M)$	$H_{\bullet, \ell}(M)$ is called <i>magnitude homology</i> [Leinster and Shulman, 2021, Definition 3.4]. The
2501.06662_F00347	20	■ 0.929	$\left(MC_{\{\bullet, \ell\}}\right)_{\ell}$	$(MC_{\bullet, \ell})_{\ell}$	because each complex $(MC_{\bullet, \ell})_{\ell}$ is bounded and moreover only finitely many ℓ can arise
2501.06662_F00348	21	■ 1.000	$q=e^{-t}$	$q = e^{-t}$	3.14. PROPOSITION. <i>For any $t > 0$, set $q = e^{-t}$. Then</i>
2501.06662_F00349	21	■ 1.000	$\bigcup_{k \geq 0} MC_{\{k, \ell\}}(\mathcal{M}) \neq$	$\bigcup_{k \geq 0} MC_{k, \ell}(\mathcal{M}) \neq$	where the first sum is over those finitely many values of $\ell \in [0, \infty)$ for which $\bigcup_{k \geq 0} MC_{k, \ell}(\mathcal{M}) \neq$
2501.06662_F00350	21	■ 1.000	$\emptyset$	$\emptyset$	$\emptyset$.
2501.06662_F00351	21	■ 1.000	$x=y_0 \rightarrow$	$x = y_0 \rightarrow$	Observe that the sum over nondegenerate paths can be restricted to those paths $x = y_0 \rightarrow$
2501.06662_F00352	21	■ 1.000	$y_1 \rightarrow \cdots \rightarrow y_k = y$	$y_1 \rightarrow \cdots \rightarrow y_k = y$	$y_1 \rightarrow \cdots \rightarrow y_k \equiv y$ such that $\prod_{i=1}^k \pi(y_i y_{i-1}) \neq 0$, which implies that $d(y_i, y_{i+1}) < \infty$ for
2501.06662_F00353	21	■ 1.000	$\prod_{i=1}^k \pi(y_i \mid y_{i-1}) \neq 0$	$\prod_{i=1}^k \pi(y_i \mid y_{i-1}) \neq 0$	$y_1 \rightarrow \cdots \rightarrow y_k = y$ such that $\prod_{i=1}^k \pi(y_i y_{i-1}) \neq 0$, which implies that $d(y_i, y_{i+1}) < \infty$ for
2501.06662_F00354	21	■ 1.000	$d(y_i, y_{i+1}) < \infty$	$d(y_i, y_{i+1}) < \infty$	$y_1 \rightarrow \cdots \rightarrow y_k = y$ such that $\prod_{i=1}^k \pi(y_i y_{i-1}) \neq 0$, which implies that $d(y_i, y_{i+1}) < \infty$ for
2501.06662_F00355	21	■ 1.000	$i=0, \ldots, k-1$	$i = 0, \dots, k-1$	any $i = 0, \dots, k-1$. Further, there is a bijective correspondence between nondegenerate
2501.06662_F00356	21	■ 1.000	$\ell := \sum_{i=0}^{k-1} d(y_i, y_{i+1}) < \infty$	$\ell := \sum_{i=0}^{k-1} d(y_i, y_{i+1}) < \infty$	paths $x = y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_k = y$ of total length $\ell := \sum_{i=0}^{k-1} d(y_i, y_{i+1}) < \infty$ and
2501.06662_F00357	21	■ 0.996	$(y_0, \dots, y_k)$	$(y_0, \dots, y_k)$	generators $(y_0, \dots, y_k)$ in $G_{k, \ell}$. Now remark that $\prod_{i=1}^k \pi(y_i y_{i-1})^t = \exp(-t\ell)$, and so it is
2501.06662_F00358	21	■ 0.996	$G_{\{k, \ell\}}$	$G_{k, \ell}$	generators $(y_0, \dots, y_k)$ in $G_{k, \ell}$. Now remark that $\prod_{i=1}^k \pi(y_i y_{i-1})^t = \exp(-t\ell)$, and so it is
2501.06662_F00359	21	■ 0.996	$\prod_{i=1}^k \pi(y_i \mid y_{i-1})^t = \exp(-t\ell)$	$\prod_{i=1}^k \pi(y_i \mid y_{i-1})^t = \exp(-t\ell)$	generators $(y_0, \dots, y_k)$ in $G_{k, \ell}$. Now remark that $\prod_{i=1}^k \pi(y_i y_{i-1})^t = \exp(-t\ell)$, and so it is
2501.06662_F00360	21	■ 0.996	$MC_{\{k, \ell\}}$	$MC_{k, \ell}$	it is important that each group $MC_{k, \ell}$ is finitely generated and that $(MC_{\bullet, \ell})_{\ell}$ is bounded. ■
2501.06662_F00361	21	■ 0.996	$\left(MC_{\{\bullet, \ell\}}\right)_{\ell}$	$(MC_{\bullet, \ell})_{\ell}$	it is important that each group $MC_{k, \ell}$ is finitely generated and that $(MC_{\bullet, \ell})_{\ell}$ is bounded. ■
2501.06662_F00362	22	■ 1.000	$H_{\{0, 0\}}(\mathcal{M})$	$H_{0, 0}(\mathcal{M})$	1. The group $H_{0, 0}(\mathcal{M})$ is the free abelian group on $\operatorname{ob}(\mathcal{M})$, and for any $\ell > 0$,
2501.06662_F00363	22	■ 1.000	$\ell > 0$	$\ell > 0$	1. The group $H_{0, 0}(\mathcal{M})$ is the free abelian group on $\operatorname{ob}(\mathcal{M})$, and for any $\ell > 0$,
2501.06662_F00364	22	■ 1.000	$H_{\{0, \ell\}}(\mathcal{M})=0$	$H_{0, \ell}(\mathcal{M}) = 0$	$H_{0, \ell}(\mathcal{M}) = 0$, cf. [Leinster and Shulman, 2021, Theorem 4.1].
2501.06662_F00365	22	■ 1.000	$H_{\{1, \ell\}}(\mathcal{M})$	$H_{1, \ell}(\mathcal{M})$	2. The group $H_{1, \ell}(\mathcal{M})$ is the free abelian group on the set of ordered pairs (x, y) such

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2501.06662_F00366	22	■ 1.000	$d(x, y)=\ell$	$d(x, y) = \ell$	that $x \neq y$ and $d(x, y) = \ell$ and there does not exist any point strictly between x
2501.06662_F00367	22	■ 1.000	$y=a_{\{0\}} a_{\{1\}} \cdots a_{\{n\}}$	$y = a_0 a_1 \cdots a_n$	elaborate, the <i>perplexity</i> of a tokenized sequence $\underline{y} = a_0 a_1 \cdots a_n$ is
2501.06662_F00368	22	■ 1.000	$p\left(-\mid y_{<i}\right)$	$p(- \mid y_{<i})$	where $p(- y_{<i})$ is the next-token probability distribution generated by the model when
2501.06662_F00369	22	■ 1.000	$y_{<i}$	$y_{<i}$	prompted with $y_{<i}$ [Jurafsky and Martin, 2025, Sec. 3.3], cf. Subsection 2.2. Perplexity
2501.06662_F00370	22	■ 1.000	$\zeta_t\left(a_{\{0\}}, y\right)$	$\zeta_t(a_0, y)$	is thus equal to the reciprocal of the zeta function $\zeta_t(a_0, y)$ when $t = 1/n$, and a_0 is the
2501.06662_F00371	22	■ 1.000	$t=1 / n$	$t = 1/n$	is thus equal to the reciprocal of the zeta function $\zeta_t(a_0, y)$ when $t = 1/n$, and a_0 is the
2501.06662_F00372	22	■ 1.000	$a_{\{0\}}$	a_0	is thus equal to the reciprocal of the zeta function $\zeta_t(a_0, y)$ when $t = 1/n$, and a_0 is the
2501.06662_F00373	22	■ 1.000	$\zeta_t\left(a_{\{0\}}, y\right)=1 / P P L(y)$	$\zeta_t(a_0, y) = 1/PPL(y)$	or said differently, $\zeta_t(a_0, y) = 1/PPL(y)$. So perhaps for intuition, one might wish to
2501.06662_F00374	22	■ 1.000	$\zeta_t(x, y)$	$\zeta_t(x, y)$	think of the zeta function $\zeta_t(x, y)$ for arbitrary x, y as a generalization of the (reciprocal)
2501.06662_F00375	22	■ 1.000	$\widehat{\mathcal{M}}:=\left[0, \infty\right]^{\mathcal{M}}$	$\widehat{\mathcal{M}} := [0, \infty]^{\mathcal{M}}$	there is a $[0, \infty]$ -category $\widehat{\mathcal{M}} := [0, \infty]^{\mathcal{M}}$ whose objects are $[0, \infty]$ -cospresheaves, that
2501.06662_F00376	22	■ 1.000	$f: \mathcal{M} \rightarrow\left[0, \infty\right]$	$f: \mathcal{M} \rightarrow [0, \infty]$	is, functions $f: \mathcal{M} \rightarrow [0, \infty]$ satisfying $\max\{f(y) - f(x), 0\} \leq \mathcal{M}(x, y)$ for all strings
2501.06662_F00377	22	■ 1.000	$\max \left\{f(y)-f(x), 0\right\} \leq \mathcal{M}(x, y)$	$\max\{f(y) - f(x), 0\} \leq \mathcal{M}(x, y)$	is, functions $f: \mathcal{M} \rightarrow [0, \infty]$ satisfying $\max\{f(y) - f(x), 0\} \leq \mathcal{M}(x, y)$ for all strings
2501.06662_F00378	22	■ 1.000	$\widehat{\mathcal{M}}$	$\widehat{\mathcal{M}}$	structure, the functor category $\widehat{\mathcal{M}}$ does, in the sense that one can compute weighted limits
2501.06662_F00379	23	■ 0.916	$p: \operatorname{ob}(\mathrm{A}) \rightarrow[0,1]$	$p: \operatorname{ob}(\mathbf{A}) \rightarrow [0, 1]$	$p: \operatorname{ob}(\mathbf{A}) \rightarrow [0, 1]$, and a similarity matrix $\theta: \operatorname{ob}(\mathbf{A}) \times \operatorname{ob}(\mathbf{A}) \rightarrow [0, \infty)$ such that
2501.06662_F00380	23	■ 0.916	$\theta: \operatorname{ob}(\mathrm{A}) \times \operatorname{ob}(\mathrm{A}) \rightarrow[0, \infty)$	$\theta: \operatorname{ob}(\mathbf{A}) \times \operatorname{ob}(\mathbf{A}) \rightarrow [0, \infty)$	$p: \operatorname{ob}(\mathbf{A}) \rightarrow [0, 1]$, and a similarity matrix $\theta: \operatorname{ob}(\mathbf{A}) \times \operatorname{ob}(\mathbf{A}) \rightarrow [0, \infty)$ such that
2501.06662_F00381	23	■ 1.000	$\theta=\zeta_t$	$\theta = \zeta_t$	In the case of a generalized metric space, $\theta = \zeta_t$ and diversity is given by
2501.06662_F00382	23	■ 1.000	$p=\pi\left(-\mid x\right)_{T(x)}$	$p = \pi(- \mid x) _{T(x)}$	work, but remark here that if we take $p = \pi(- x) _{T(x)}$, then we recover the Shannon
2501.06662_F00383	23	■ 0.999	$\pi\left(-\mid x\right)_{T(x)}$	$\pi(- \mid x) _{T(x)}$	entropy of $\pi(- x) _{T(x)}$, which unlike our expression for the magnitude does not treat all
2501.06662_F00384	23	■ 1.000	$P: A \times A \rightarrow[0,1]$	$P: A \times A \rightarrow [0, 1]$	chain with transition matrix $\underline{P}: A \times A \rightarrow [0, 1]$ and with stationary distribution $p_\perp$, in
2501.06662_F00385	23	■ 1.000	$p_{\{\perp\}}$	$p_\perp$	chain with transition matrix $P: A \times A \rightarrow [0, 1]$ and with stationary distribution $p_\perp$, in
2501.06662_F00386	24	■ 1.000	$\{ \}^{\sim}$	$\sim$	with language models. https://web.stanford.edu/~jurafsky/slp3/ , 2025.